

Chapter 8: Diffusion MRI

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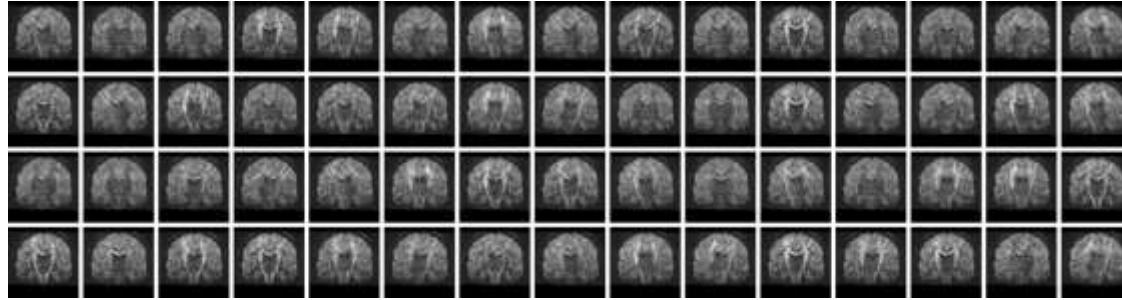
January 12/19, 2026

Image Acquisition and Analysis in Neuroscience

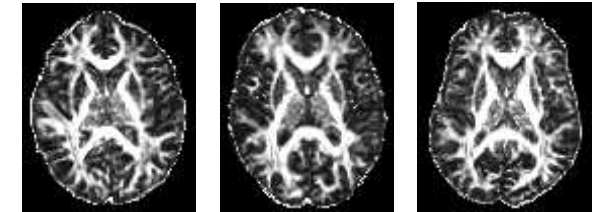
Acquisition



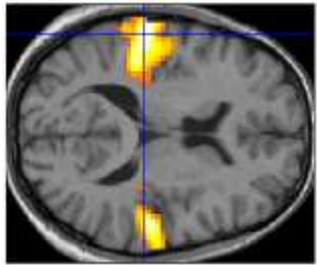
Reconstruction & Artifact Correction



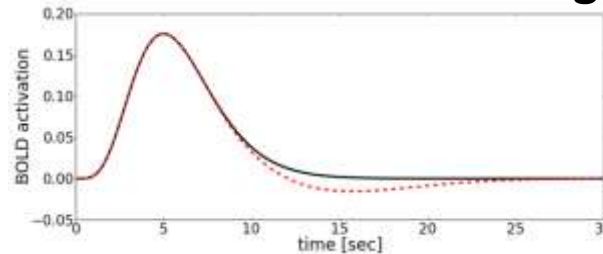
Registration



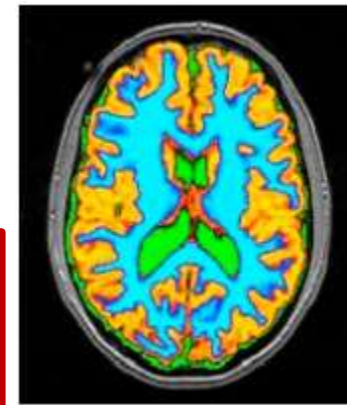
Hypothesis Testing



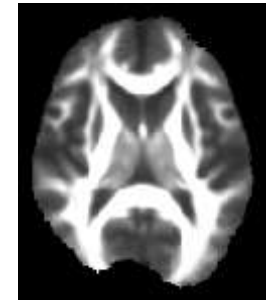
Mathematical Modeling



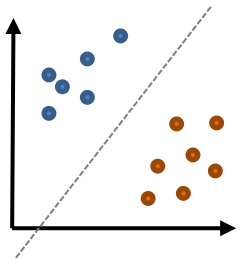
Segmentation



Atlas Construction



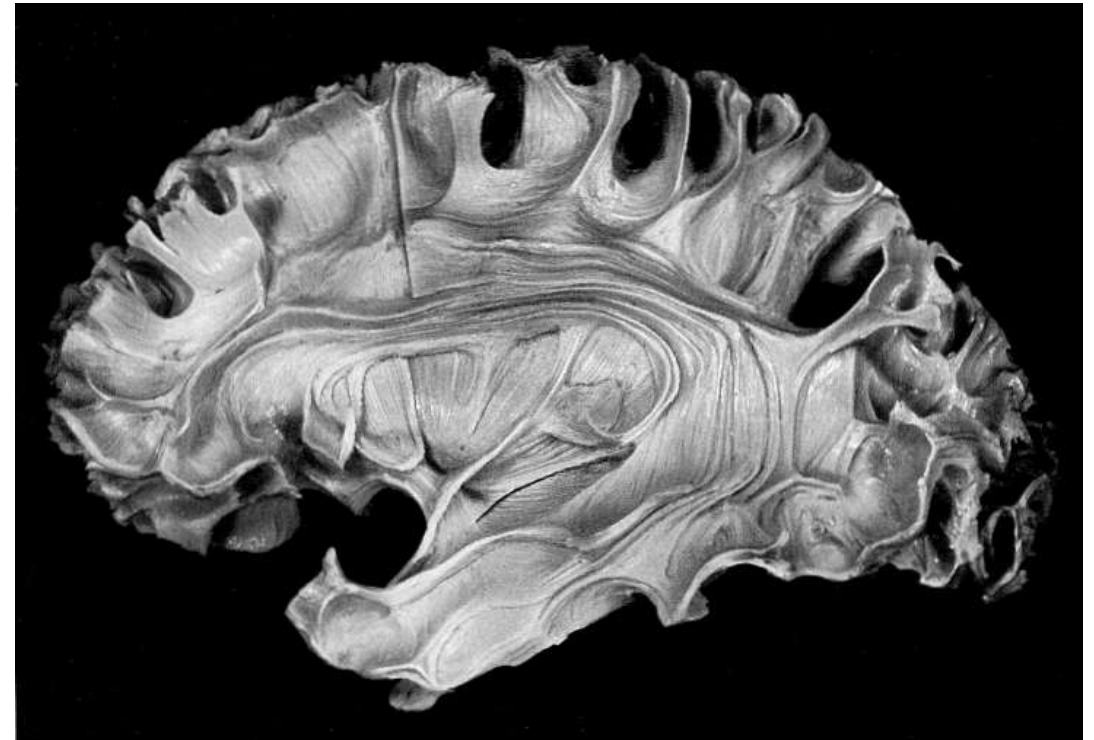
Machine Learning



8.1 Diffusion Tensor MRI

Background: Klingler Dissection Method

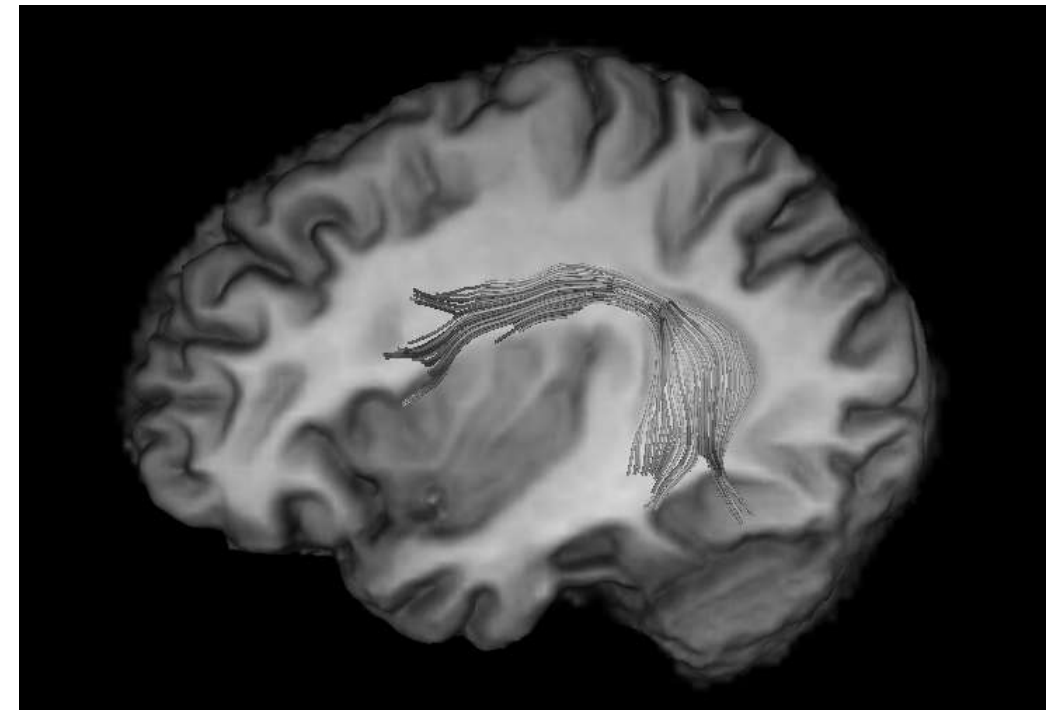
- **Klingler dissection** is a traditional way of visualizing white matter tracts
 - Developed by Joseph Klingler in the 1930s
 - Formalin-fixing and freezing the brain before scratching away tissue with a blunt instrument
 - Reveals fiber tracts in their natural anatomical context



Photograph of a Klingler Dissection

Virtual Klingler Dissection via Diffusion Tensor MRI

- **Diffusion MRI** permits the non-invasive *in vivo* visualization of white matter tracts
 - **Spatial trajectories** algorithmically reconstructed based on diffusion measurements
 - Measurements also reflect local **fiber density and myelination**, can be used to study brain maturation or disease
- *Image*: Joint visualization of traditional and diffusion MRI that has been inspired by Klingler Dissection



[Schultz et al., CGF 2008]

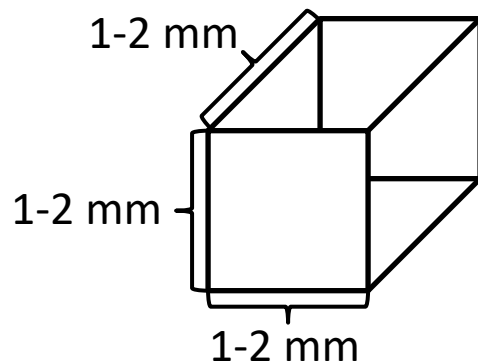
Introduction to Diffusion MRI

Goal: Investigate the microstructure of biological tissue using Magnetic Resonance Imaging (MRI)

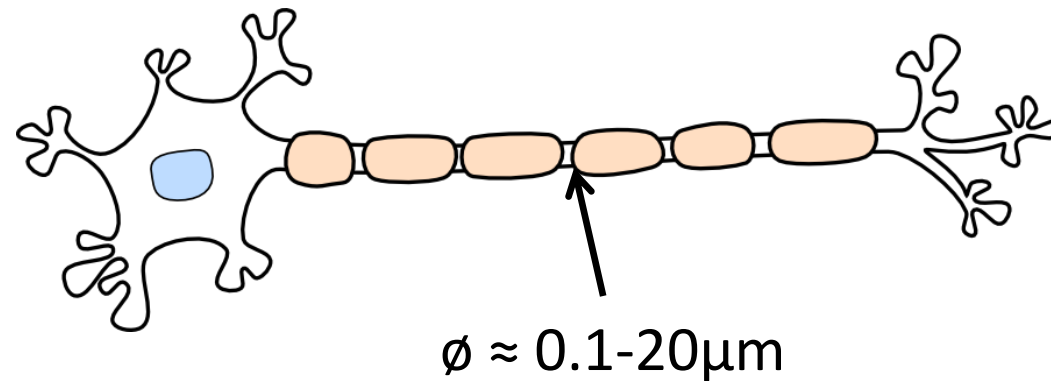
- Most frequently: White matter tracts in the brain



Challenge: Voxel size is far too large to resolve the structures of interest



Voxel size



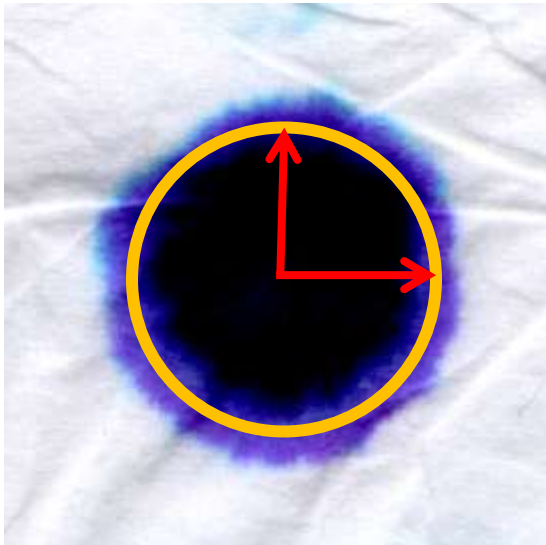
Axon (nerve fiber) size

Introduction to Diffusion MRI

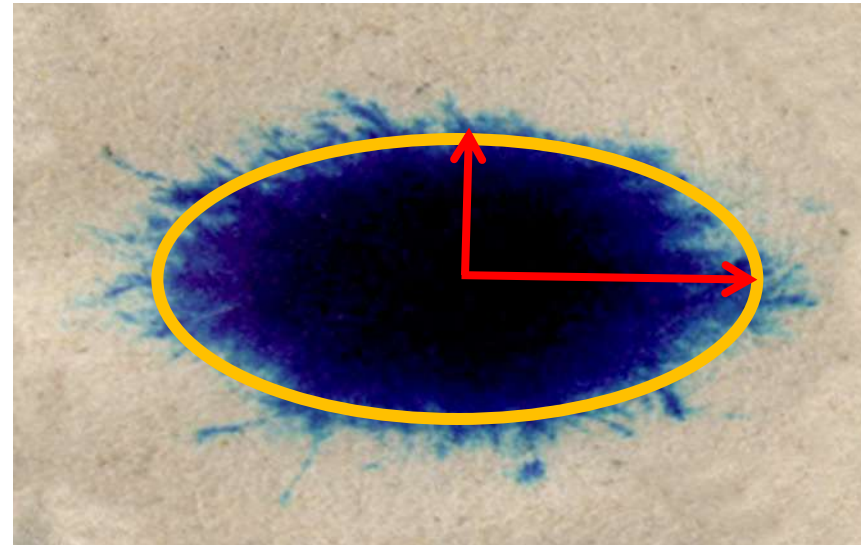
Approach: Use water molecules as a contrast agent

- Exploits their spontaneous heat motion at the desired spatial scale

Analogy: Observe diffusion of ink on paper



Kleenex



Newspaper

Brownian Motion and Diffusivities

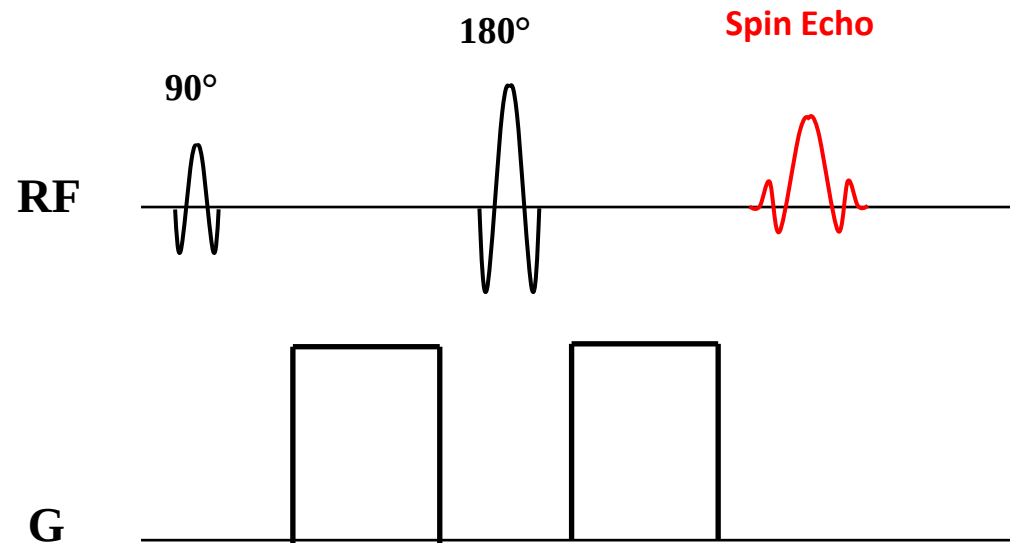
Molecular diffusion = Brownian motion of water molecules

- Random walk due to thermal energy
- Measured with the **Diffusion Coefficient D**
 - Proportionality constant in Fick's law:
$$J = -D\nabla c$$
 - Brownian motion happens even without a concentration gradient („self-diffusion“)
- In white matter: $D \approx 1-5 \cdot 10^{-4} \text{ mm}^2/\text{s}$
- Due to interactions with cellular structures, D depends on spatial direction (anisotropic diffusion)

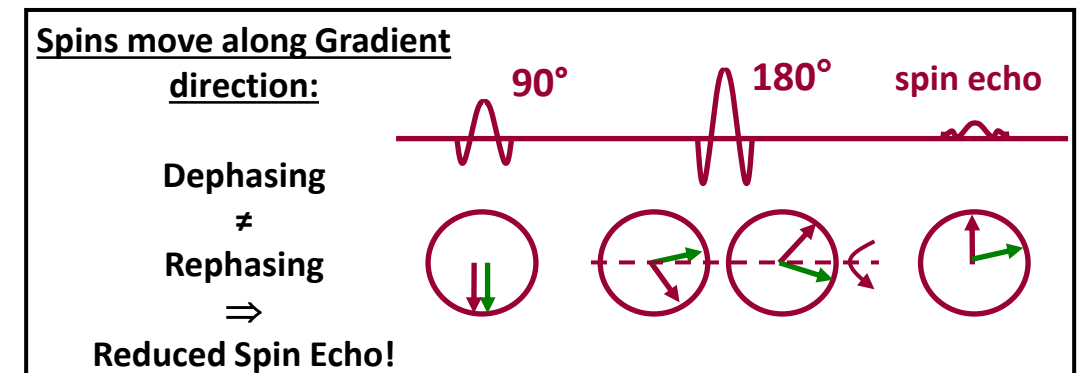
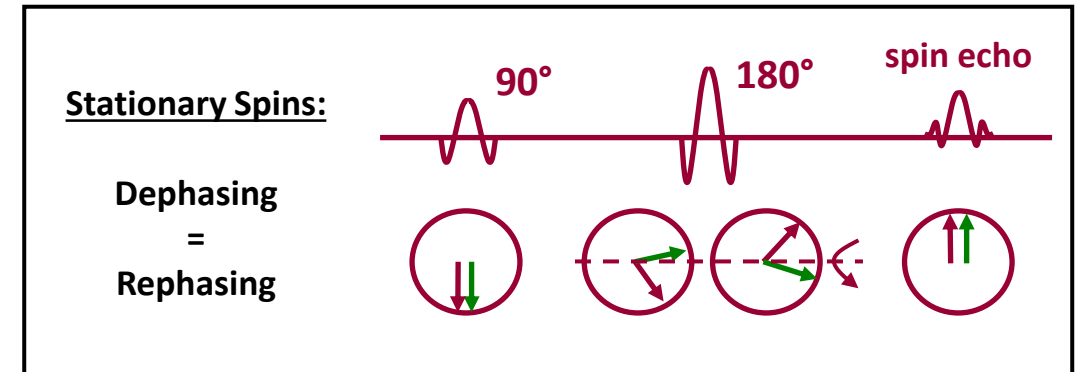


Stejskal-Tanner MR Sequence

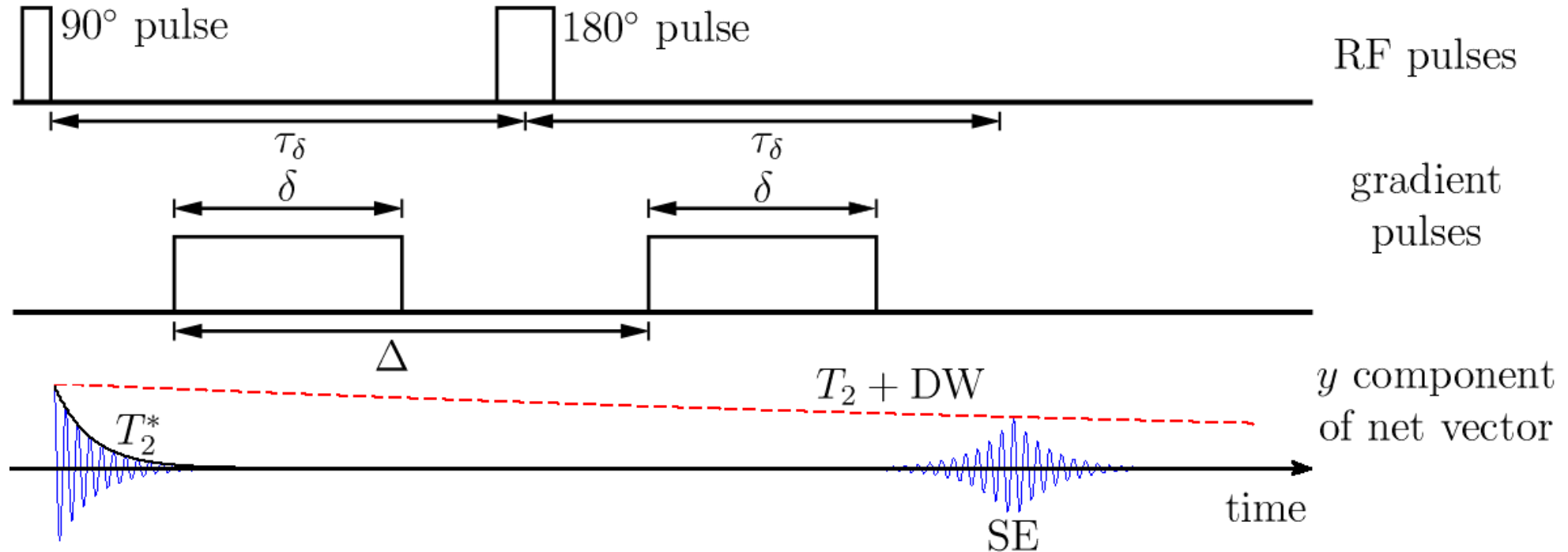
Stejskal-Tanner Sequence modifies Spin Echo to introduce a diffusion weighting



Diffusion weighting with gradients:
De- and re-phasing of spins depend on their spatial position.



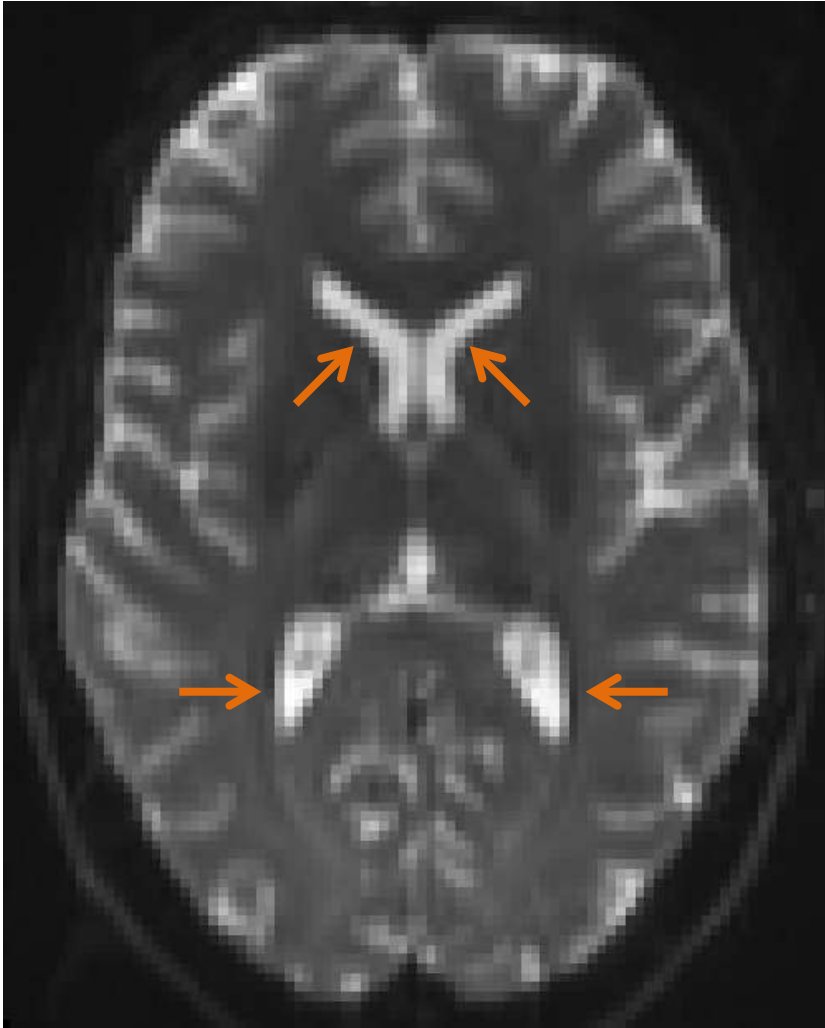
Definition of the b Value



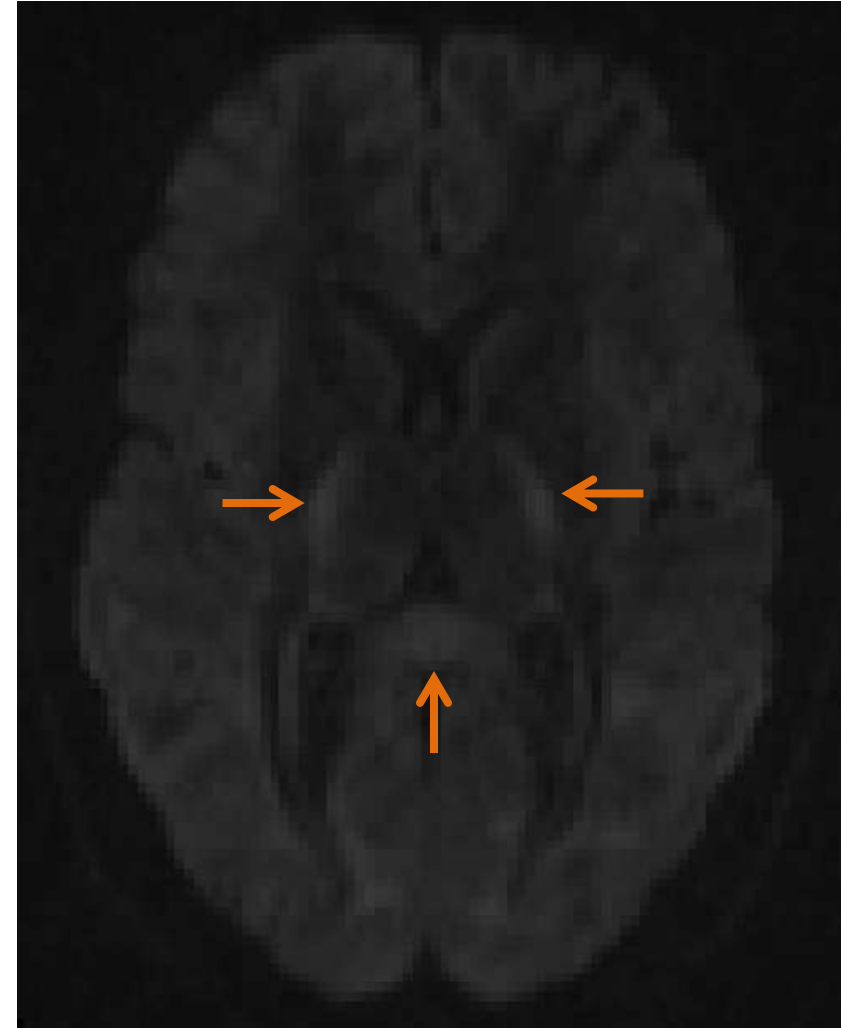
Signal model for the Stejskal-Tanner pulse sequence:

$$\frac{S(g, \delta, \Delta)}{S_0} = \exp\left(-\gamma^2 g^2 \delta^2 \left(\Delta - \frac{\delta}{3}\right) D\right) \equiv \exp(-bD)$$

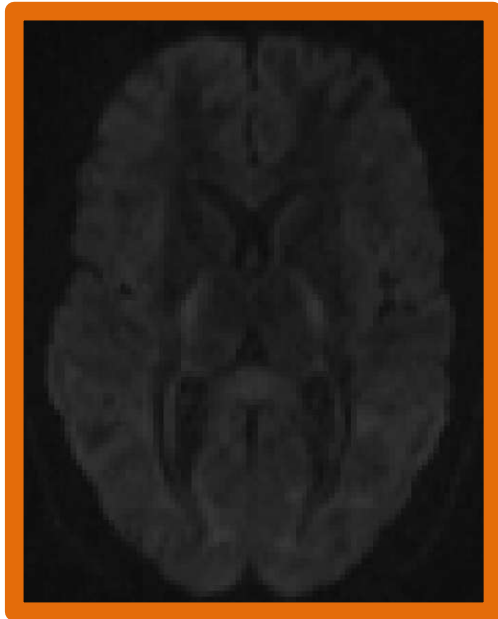
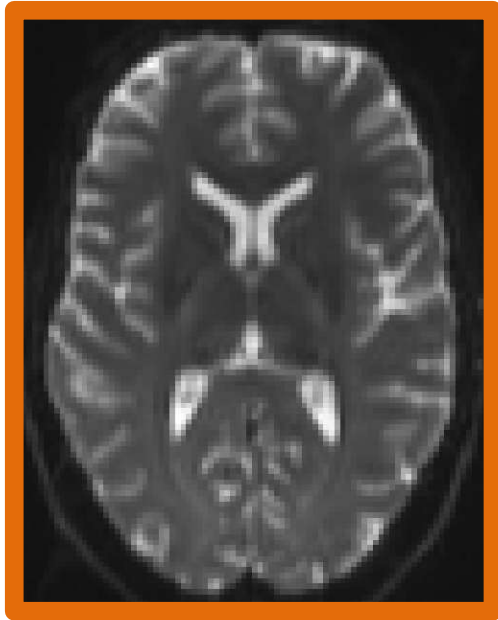
Illustration of Diffusion Weighting



Standard T₂ MRI



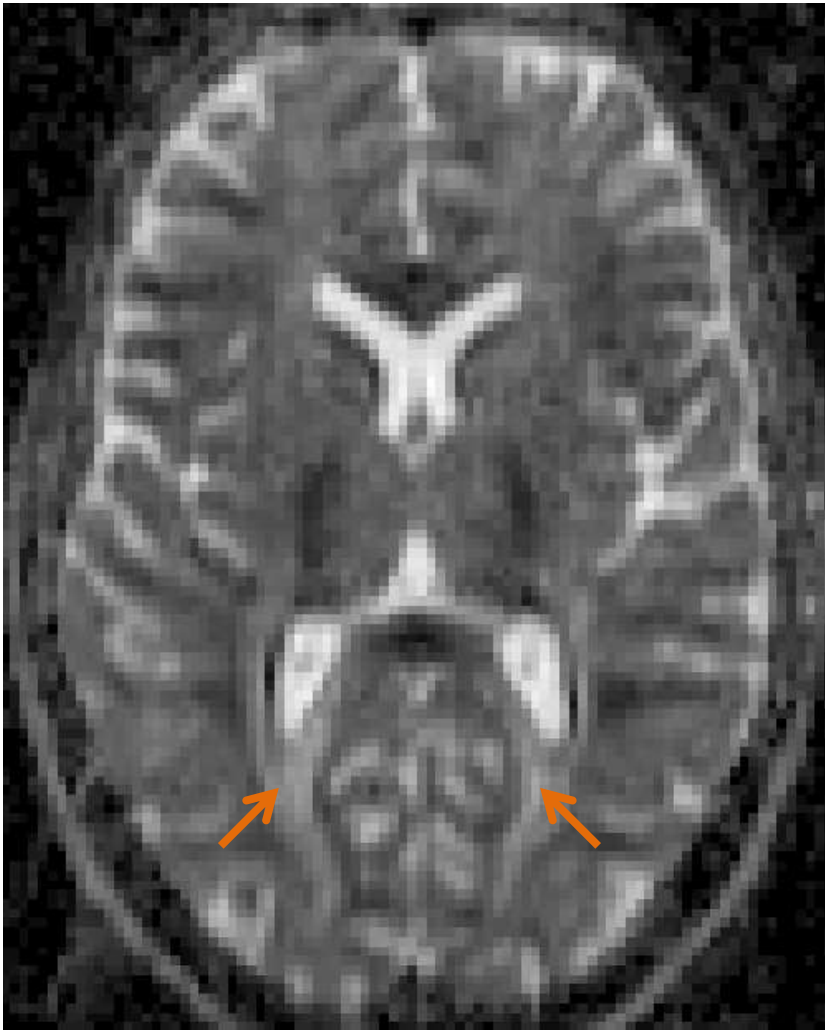
Diffusion Weighted MRI



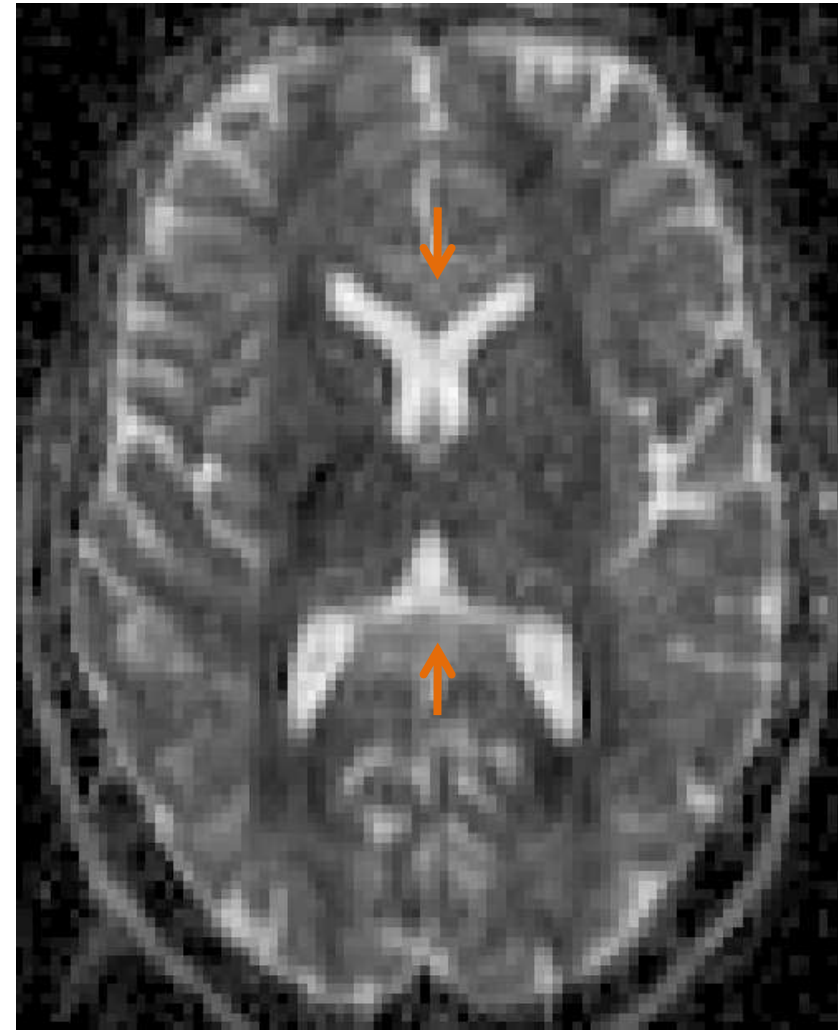
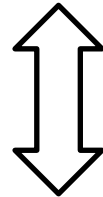
Stejskal-Tanner Equation:

$$S(d) = S_0 e^{-bd}$$

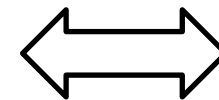


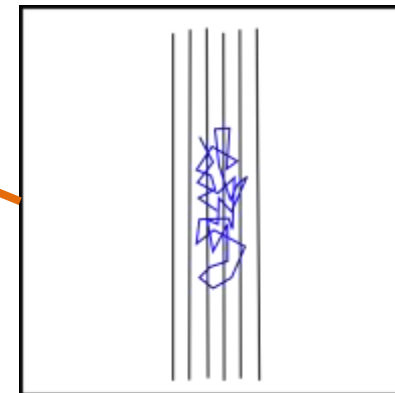
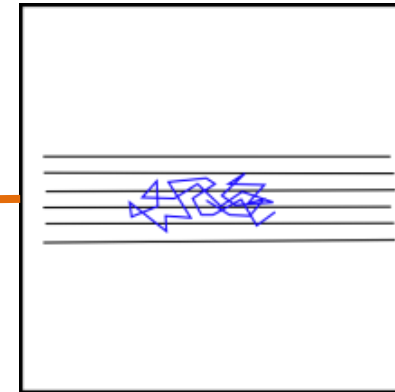
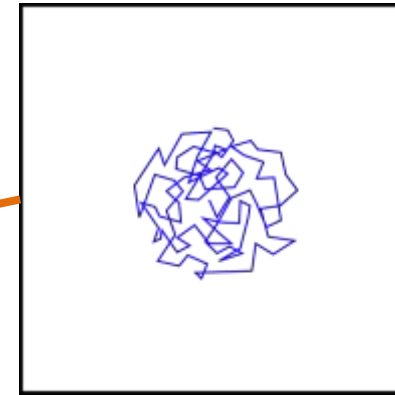
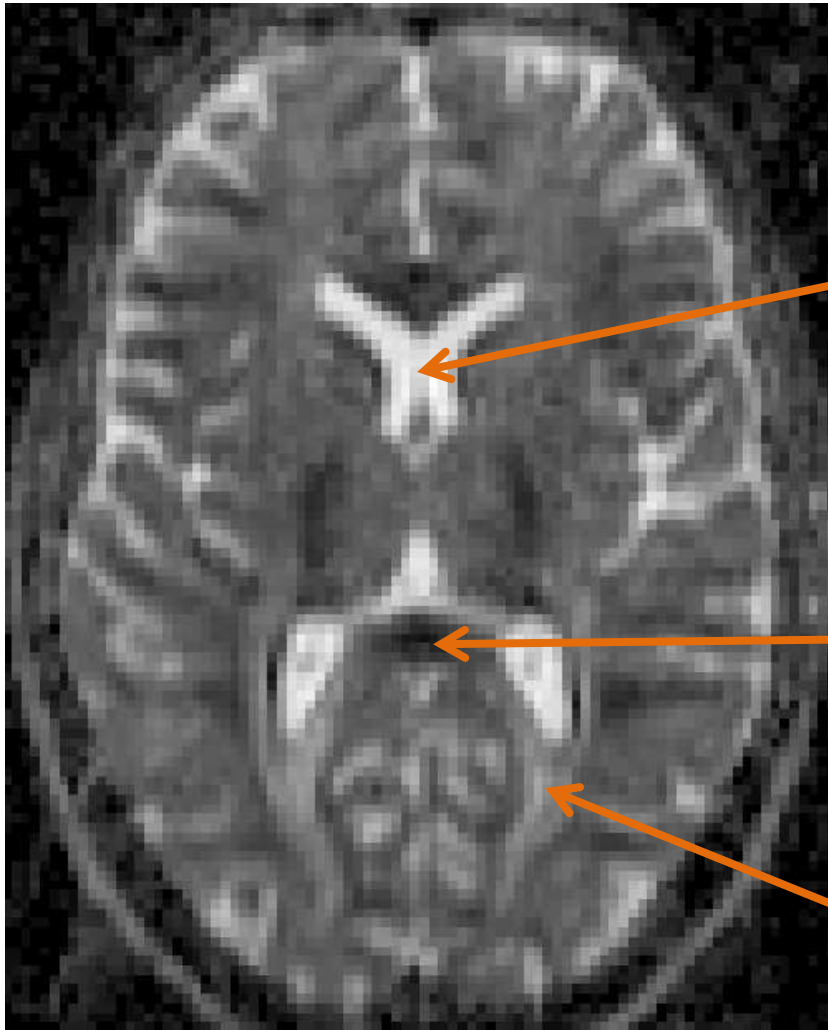


Vertical Gradient

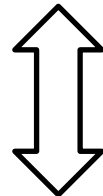


Horizontal Gradient





Vertical Gradient



The Diffusion Tensor Model

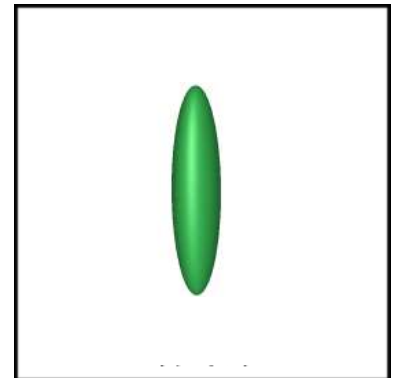
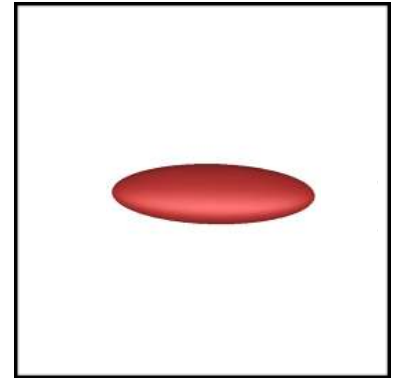
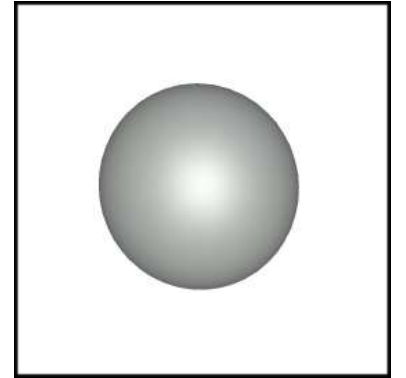
- **Diffusion Tensor MRI** models diffusivity D as a quadratic function of gradient direction $\mathbf{x}$:

$$S(D(\mathbf{x})) = S_0 e^{-bD(\mathbf{x})}$$

with

$$D(\mathbf{x}) = \mathbf{x}^T \mathbf{D} \mathbf{x}$$

- The **diffusion tensor** $\mathbf{D}$, a 3×3 symmetric matrix, contains the coefficients of the diffusivity function



Eigenvector Decomposition

Diffusion tensor **D** decomposes into:

- 3 eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$
 - If **D** is positive definite: All $\lambda > 0$
 - *Note:* Diffusivity is a non-negative physical quantity.
Due to measurement noise, we might obtain $S_0 < S(d)$, which can lead to **D** with negative λ 's
- 3 orthogonal eigenvectors $\mathbf{e}_1/\mathbf{e}_2/\mathbf{e}_3$
 - $\mathbf{e}_1$ indicates single main fiber orientation

Axes of **tensor ellipsoid** are

- aligned with eigenvectors and
- scaled by eigenvalues



Estimating Diffusion Tensors

- **Stejskal-Tanner Equation:**

$$S(\mathbf{x}) = S_0 e^{-b\mathbf{x}^T \mathbf{D} \mathbf{x}}$$

- Taking the logarithm and solving for $\mathbf{D}$ produces a system of linear equations (one per measurement i):

$$\underbrace{\sum_{k,l} [\mathbf{D}]_{kl} [\mathbf{x}_i]_k [\mathbf{x}_i]_l}_{\mathbf{x}_i^T \mathbf{D} \mathbf{x}_i} = -\frac{1}{b} \ln \frac{S(\mathbf{x}_i)}{S_0}$$

- $\mathbf{D}$ = symmetric 3x3 matrix $\rightarrow$ six free variables
 - Due to noise, we usually take more than six measurements and find (weighted) least squares solution

Mean Diffusivity

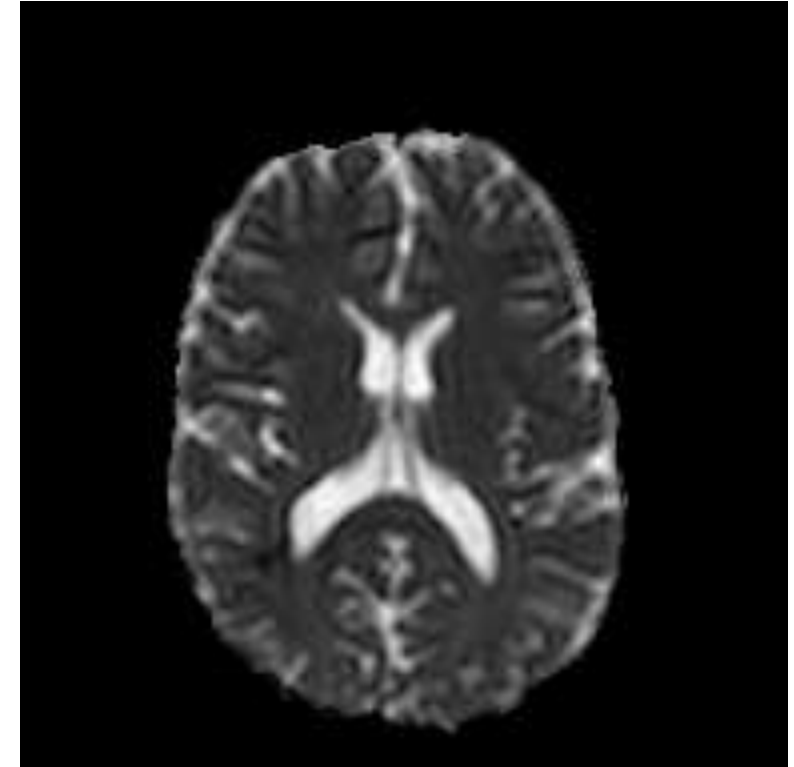
- **Mean Diffusivity (MD)**

- Average eigenvalue: $(\lambda_1 + \lambda_2 + \lambda_3)/3$

- Interpretation:

- Average diffusivity (over all directions)

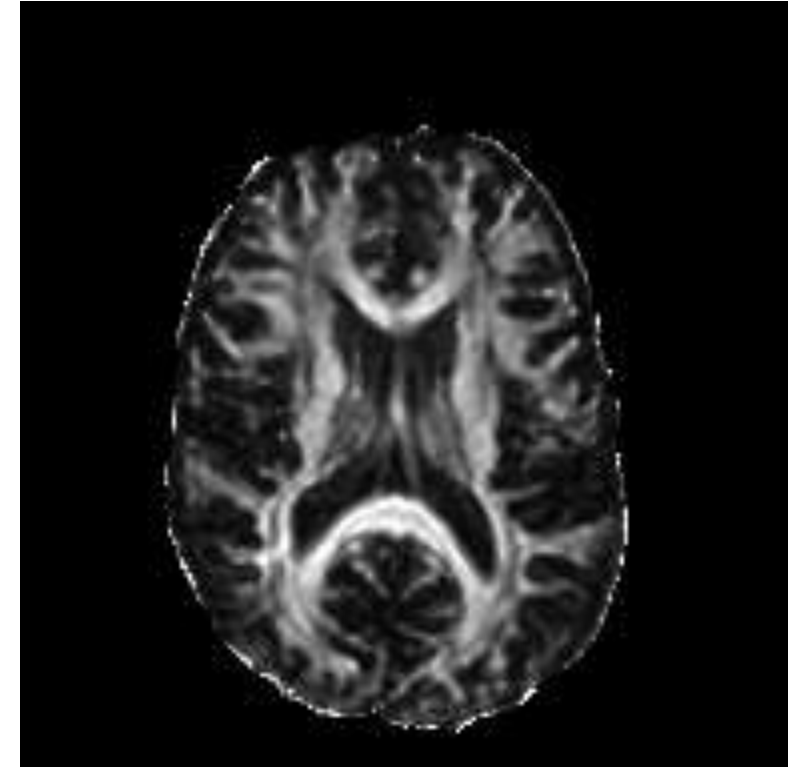
- MD much higher in ventricles than in tissue
 - Approx. constant in healthy tissue
 - Clinically used as an early indicator of stroke
 - Elevations can indicate a loss of white matter integrity



Fractional Anisotropy (FA)

- **Fractional Anisotropy**

- Quantifies the degree of anisotropy
 - High values in white matter
 - Correlates with fiber density / integrity
 - Also correlates with orientation dispersion
- Based on variance of eigenvalues
- Normalized to [0,1]
(if positive definite)



$$FA = \sqrt{\frac{3}{2}} \frac{\sqrt{(\lambda_1 - MD)^2 + (\lambda_2 - MD)^2 + (\lambda_3 - MD)^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

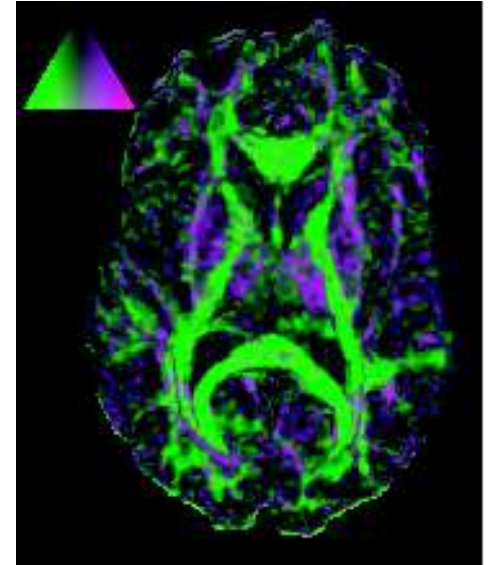
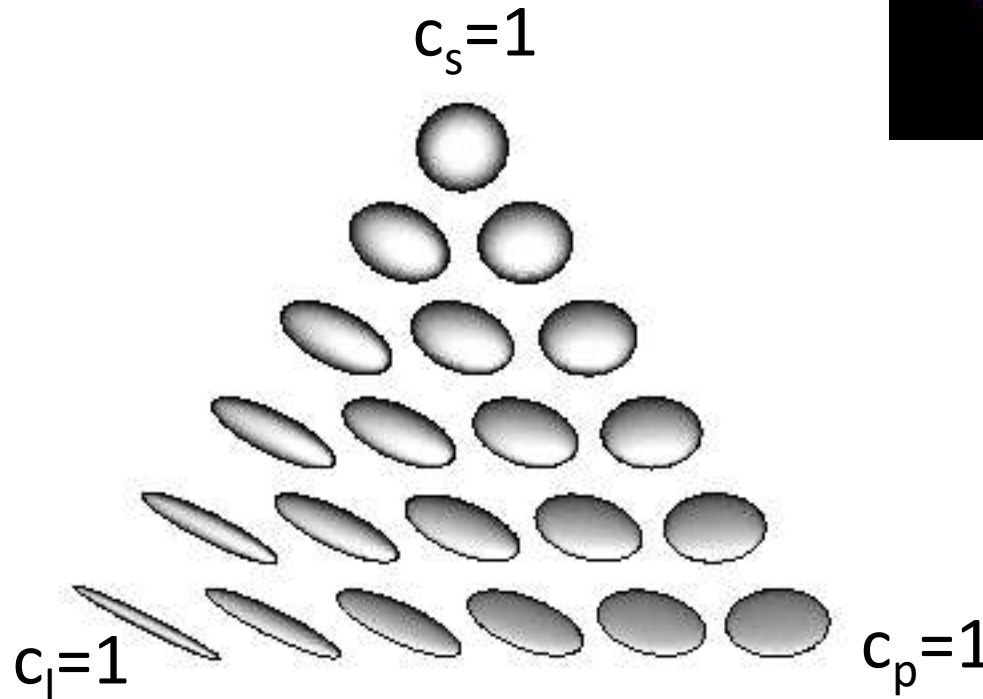
Westin Measures

- Westin's c_l , c_p , c_s quantify the extent to which the tensor ellipsoid is linear / planar / spherical
 - Based on sorted eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$

$$c_l = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3}$$

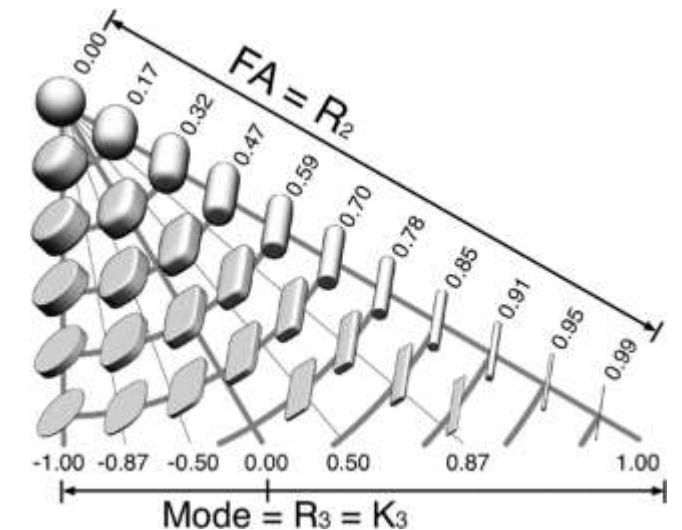
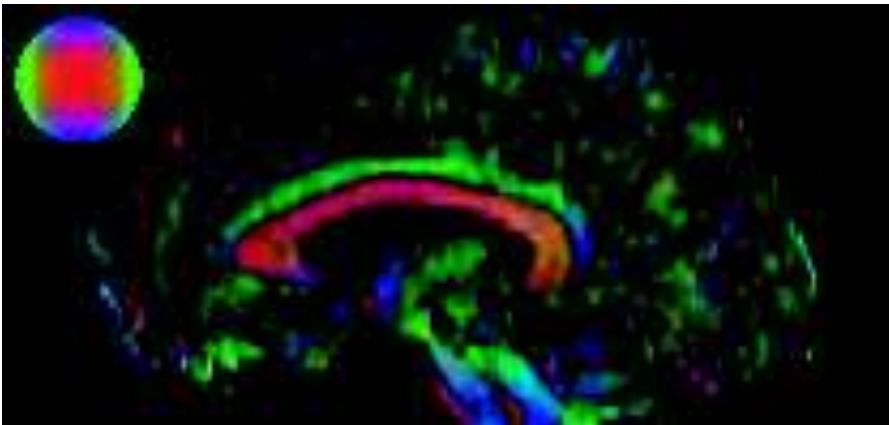
$$c_p = \frac{2(\lambda_2 - \lambda_3)}{\lambda_1 + \lambda_2 + \lambda_3}$$

$$c_s = \frac{3\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3}$$



Color Coded Eigenvector Maps

- Standard Color Coding: XYZ->RGB
- Usually applied to principal eigenvector:
$$R = |v_{1x}|, \quad G = |v_{1y}|, \quad B = |v_{1z}|$$
- Usually modulated by FA
 - c_i makes more sense; FA can be large even for planar anisotropy!



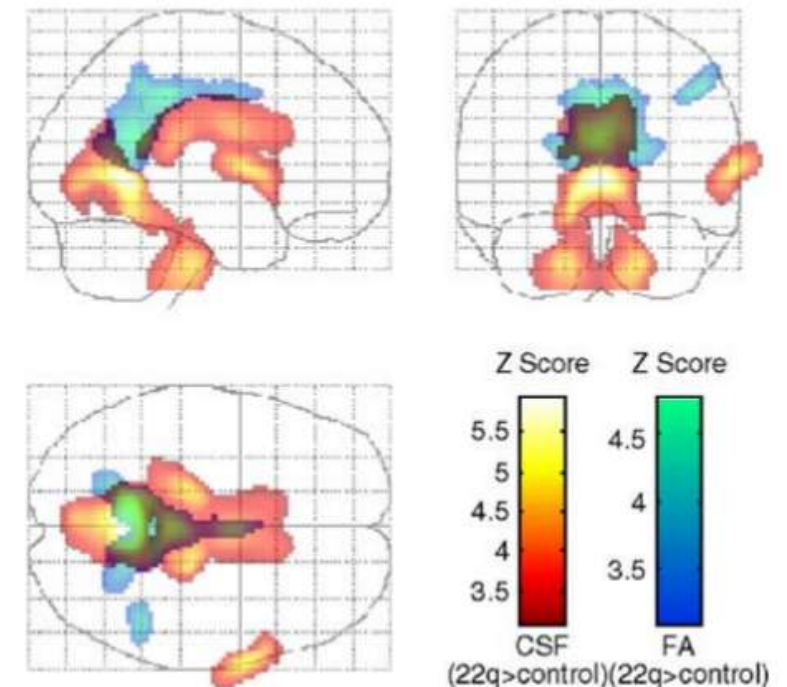
Summary: Diffusion Tensor MRI

- DT-MRI is based on measuring the magnitude of **Brownian heat motion**
 - To a certain extent, DT-MRI is **quantitative** (calibrated)
- **Probe for tissue microstructure:** Interactions between water and tissue leave a “footprint” on diffusion behavior
 - Exploring it requires many measurements, 5 minutes or more scan time for full-brain analysis
- **Model:** Exponential decay with quadratic diffusivity
 - Diffusion Tensor = Coefficients of the quadratic form
 - Can be thought of in terms of diffusion ellipsoid
 - Main parameters for statistical analysis: Fractional Anisotropy (FA), Mean Diffusivity (MD)

8.2 Tract-Based Spatial Statistics

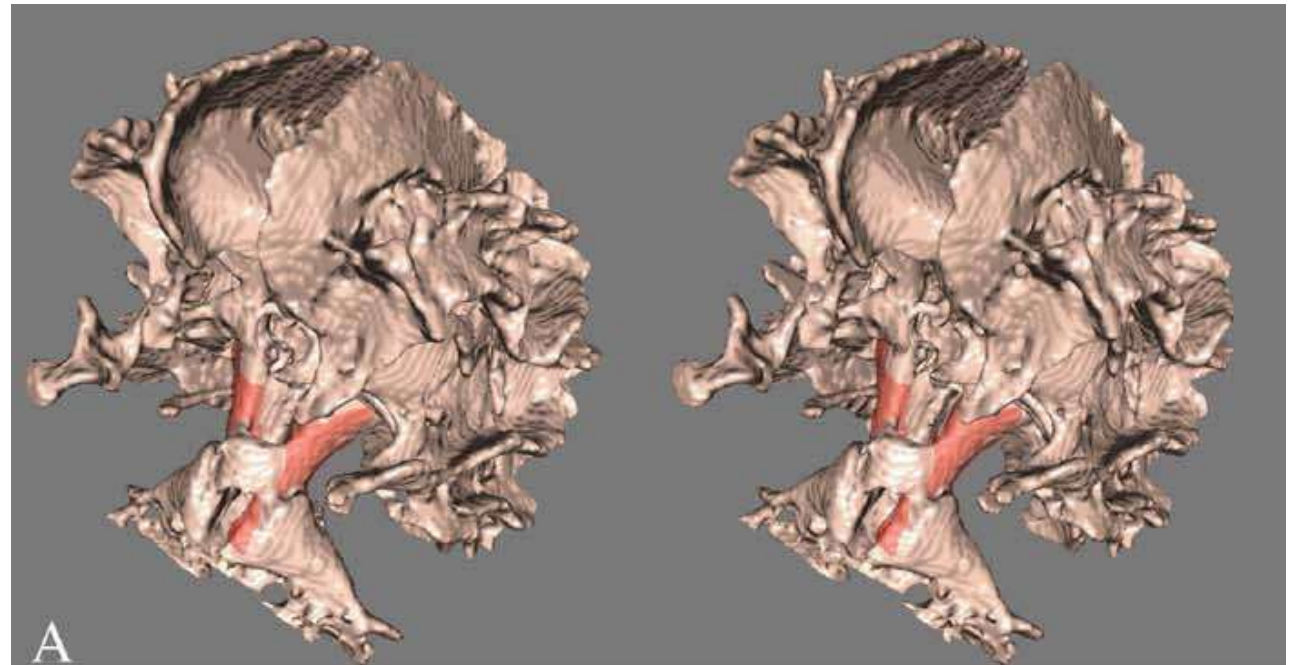
Motivation

- **Natural approach** to full-brain analysis of Fractional Anisotropy maps:
 - Normalization + Smoothing + Statistical Parametric Mapping
- **Problems:**
 - FA varies strongly perpendicular to white matter tracts
 - This makes results difficult to interpret if alignment is imperfect
 - Would like to avoid smoothing, since it works against our goal of breaking the resolution limit of MRI

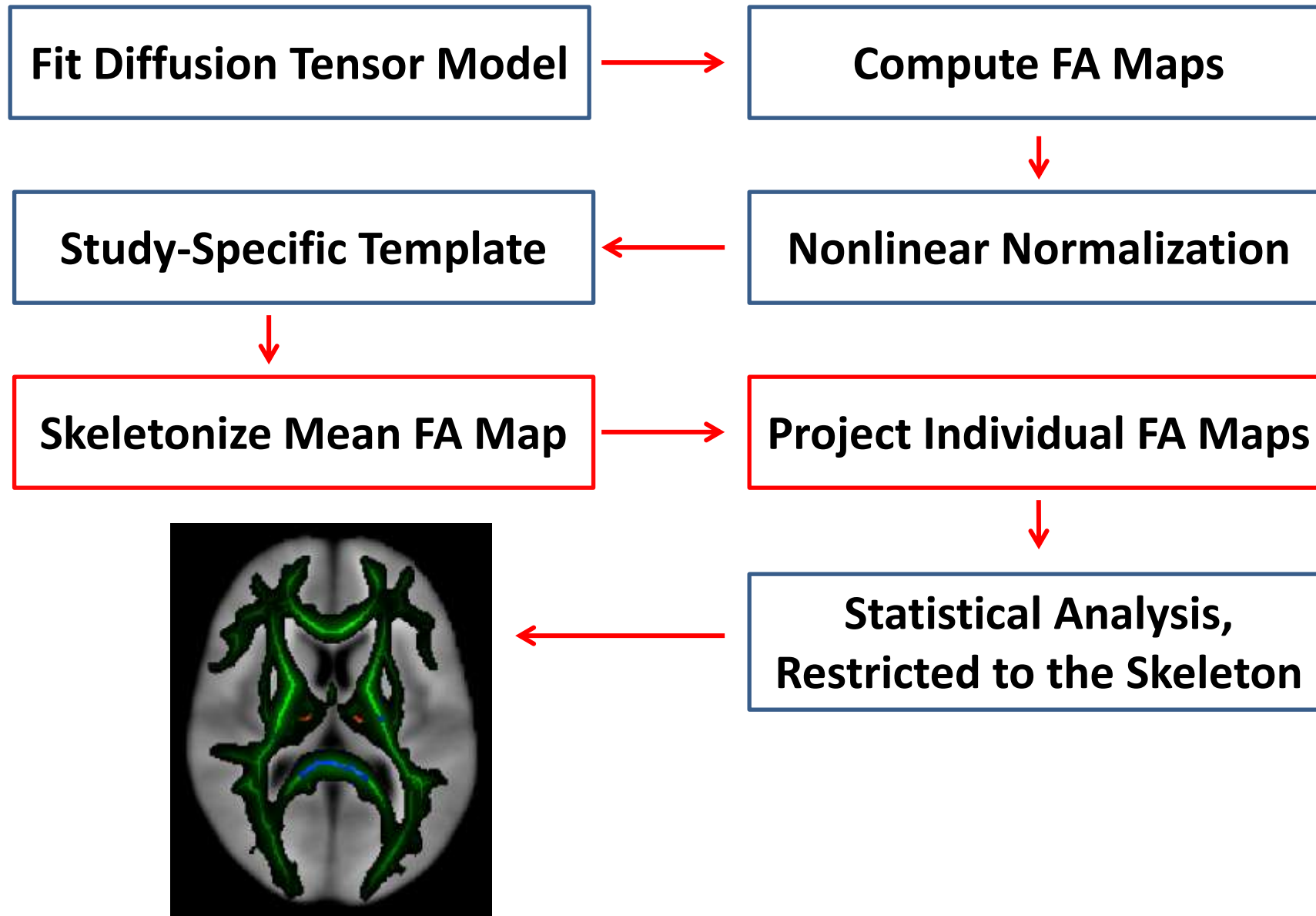


Tract-Based Spatial Statistics (TBSS)

- **Idea:** Only consider values along the center (“skeleton”) of white matter structures
 - Center characterized by locally maximal Fractional Anisotropy
 - TBSS computes a medial surface in the group average FA map, then projects largest FA values from all subjects to it
 - Compensates residual misalignment *without any smoothing*
 - Also reduces the family-wise error problem

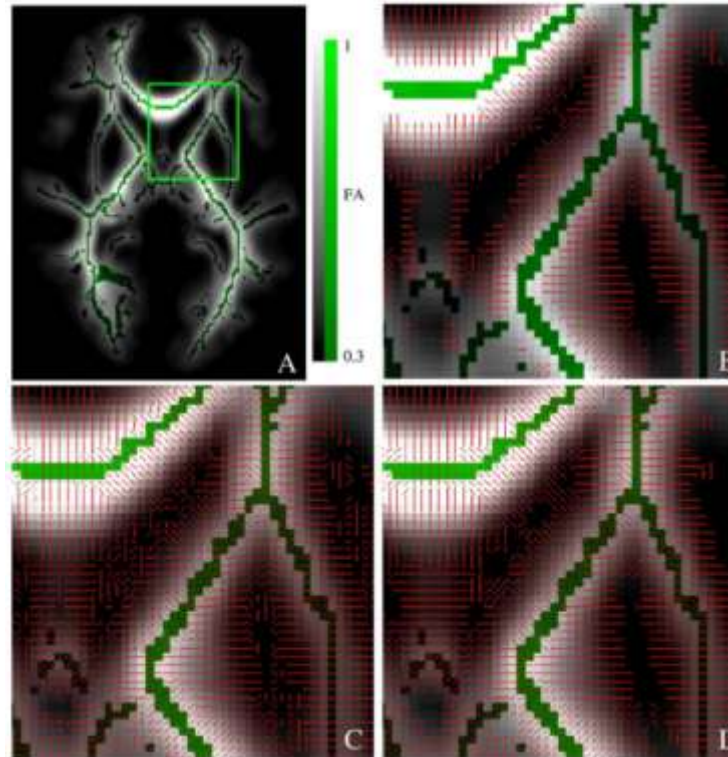


The TBSS Pipeline



Mean FA Skeletonization

- Construct a quantized vector field perpendicular to the skeleton:



1. Gradient direction $\mathbf{g}$ (where strong enough)

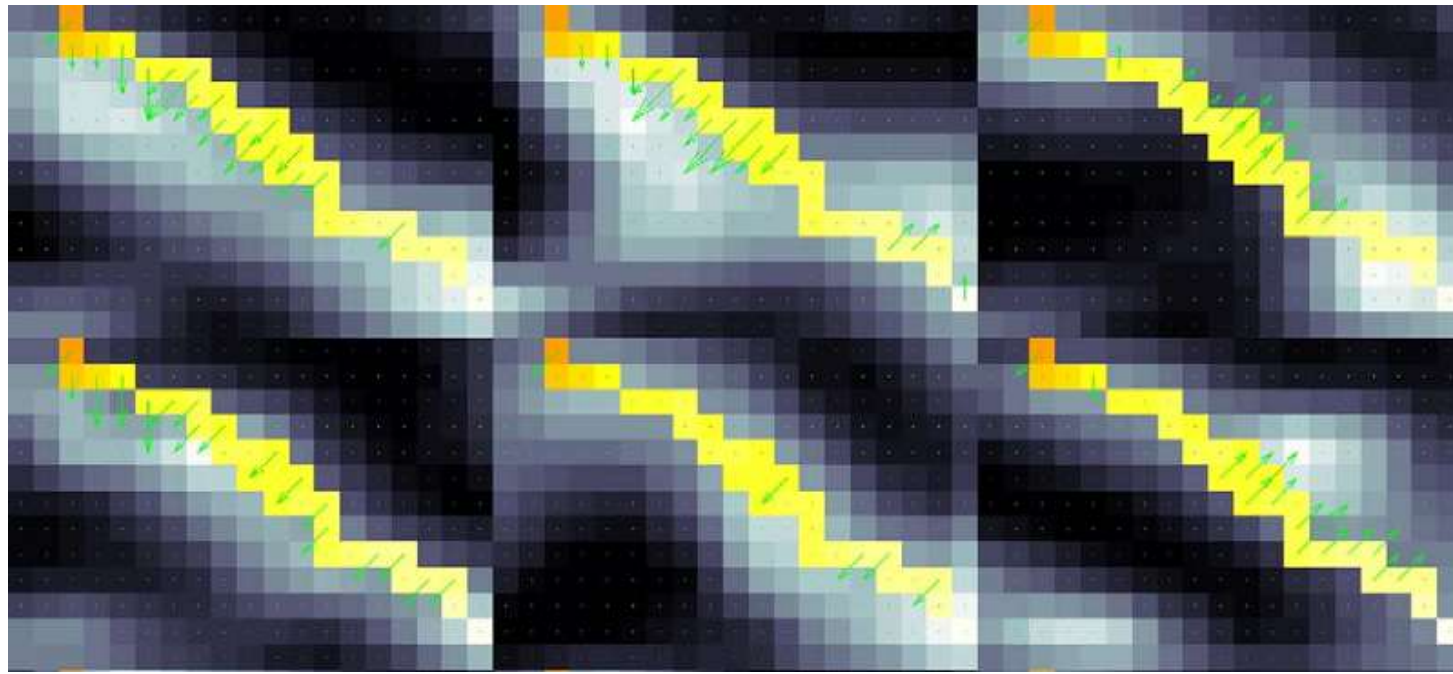
2. Direction $\mathbf{e}_3$ of strongest concavity (elsewhere)

3. Slight smoothing

- Define skeleton as voxels at which FA is locally maximal in direction of that vector field
- Discard voxels with low FA (e.g., <0.2)

Projection to the Skeleton

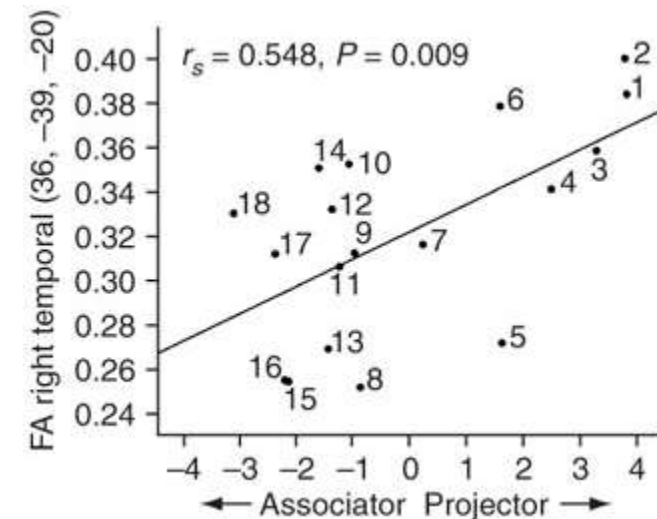
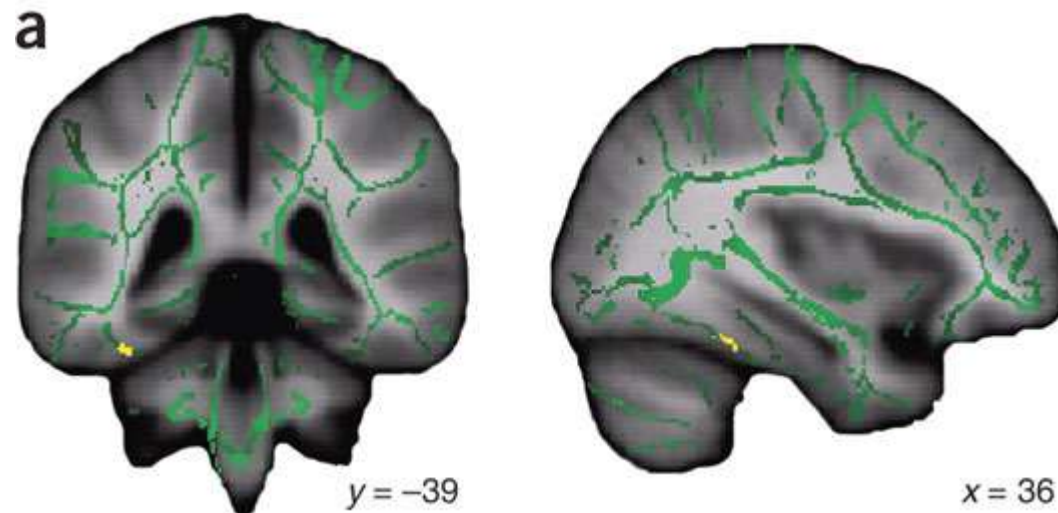
- Find **maximum FA** along the vector field used before
 - Ensure each voxel can only project to one skeleton point
 - Impose a soft distance limit



In images, skeleton is yellow, arrows point to voxel *from which value was taken*

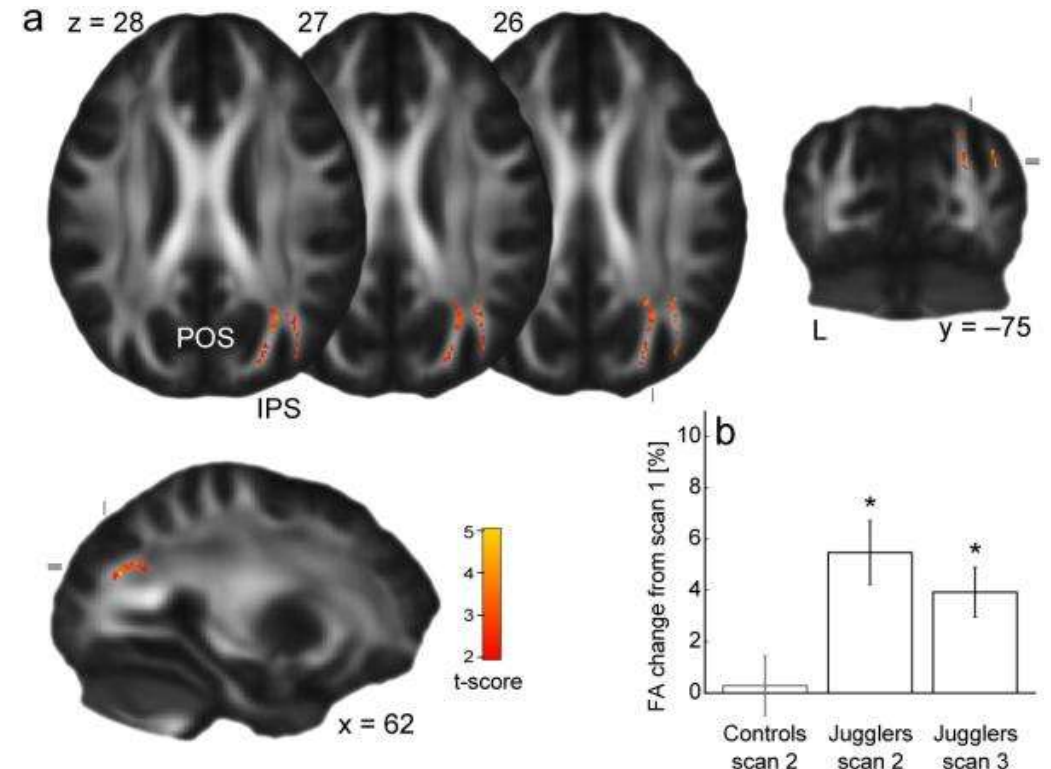
TBSS: Synesthesia

- [Rouw/Scholte 2007] showed that **grapheme-color synesthesia** is associated with increased FA in several white matter regions
 - 18 synesthetes vs. 18 normal controls
 - FA in fusiform gyrus correlated with “strength” of synesthetic experience



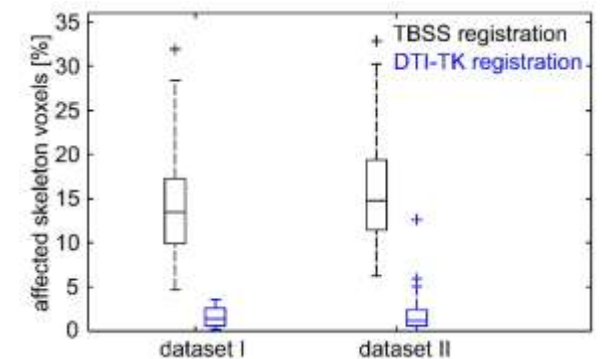
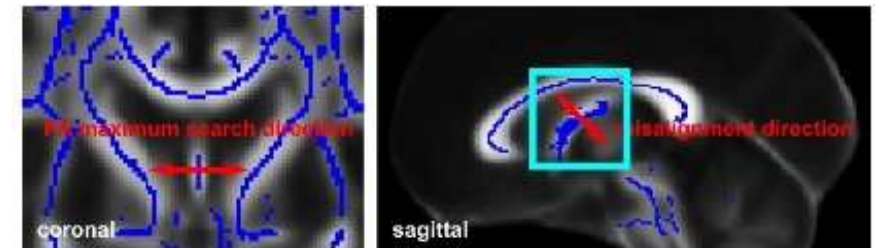
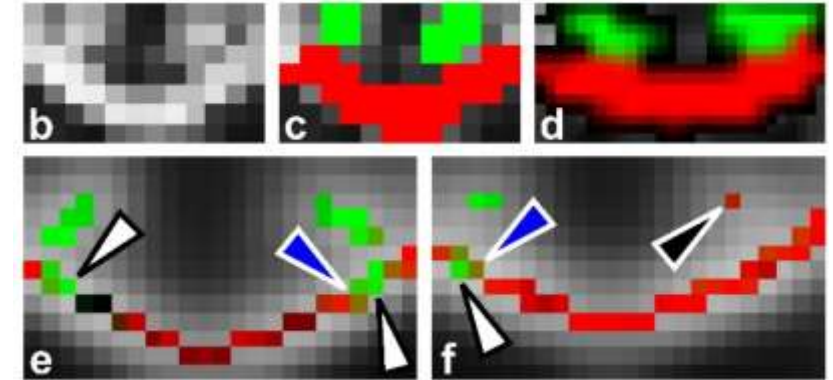
TBSS: A Juggling Study

- [Scholz et al. 2009] showed that **learning how to juggle** not only leads to observable changes in gray matter density, but also to increased FA in adjacent white matter
 - 48 subjects assigned to equally sized training and control groups
 - Longitudinal comparison (across time)



Criticism of TBSS

- TBSS projection is known not to be 100% reliable:
 - Voxels from nearby bundles might be assigned to an incorrect part of the skeleton
 - Projection might fail to compensate misalignment
 - Results depend on choice of atlas
- *Recommendations:*
 - Normalize based on full tensor information
 - Verify findings via back-projection



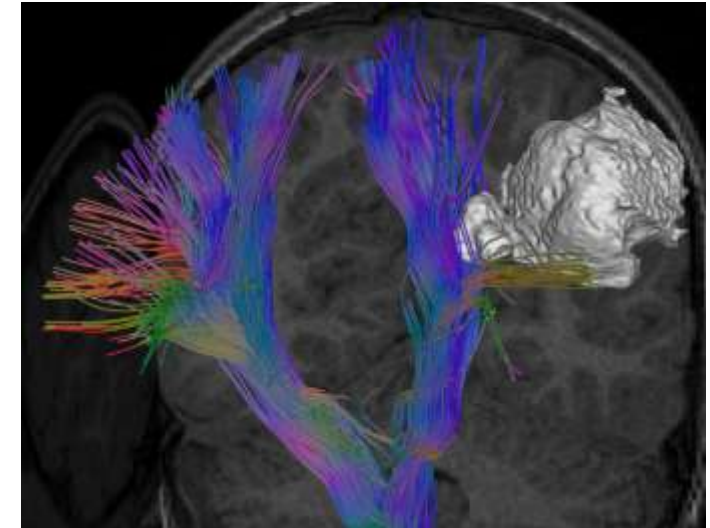
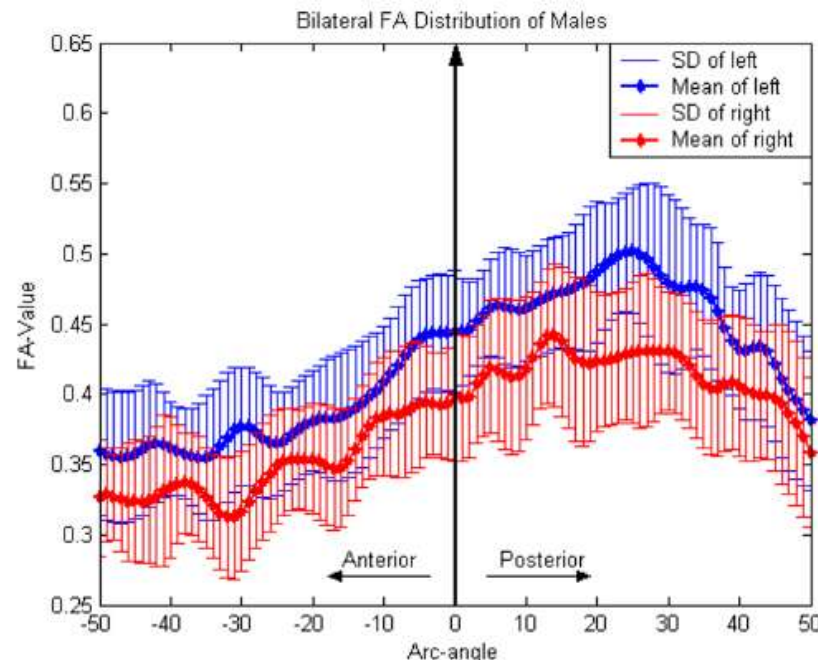
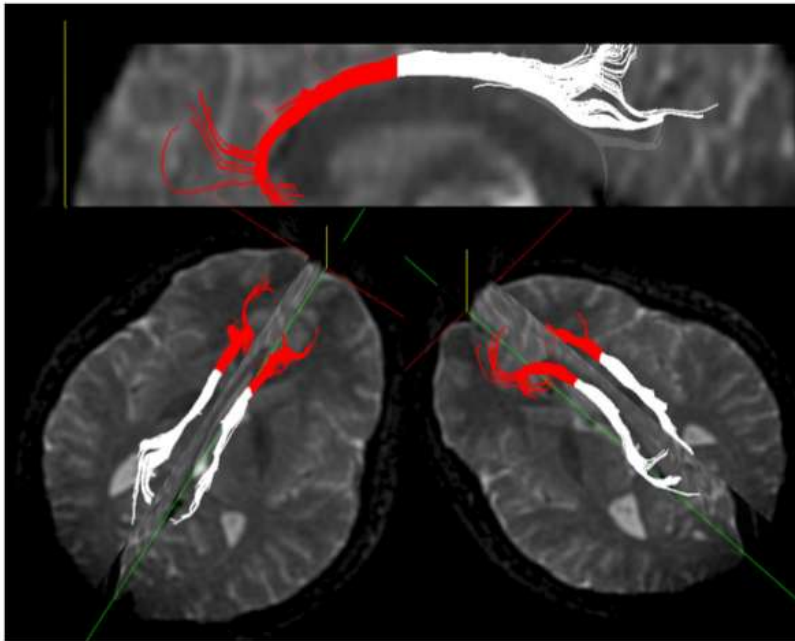
Summary: Tract-Based Spatial Statistics

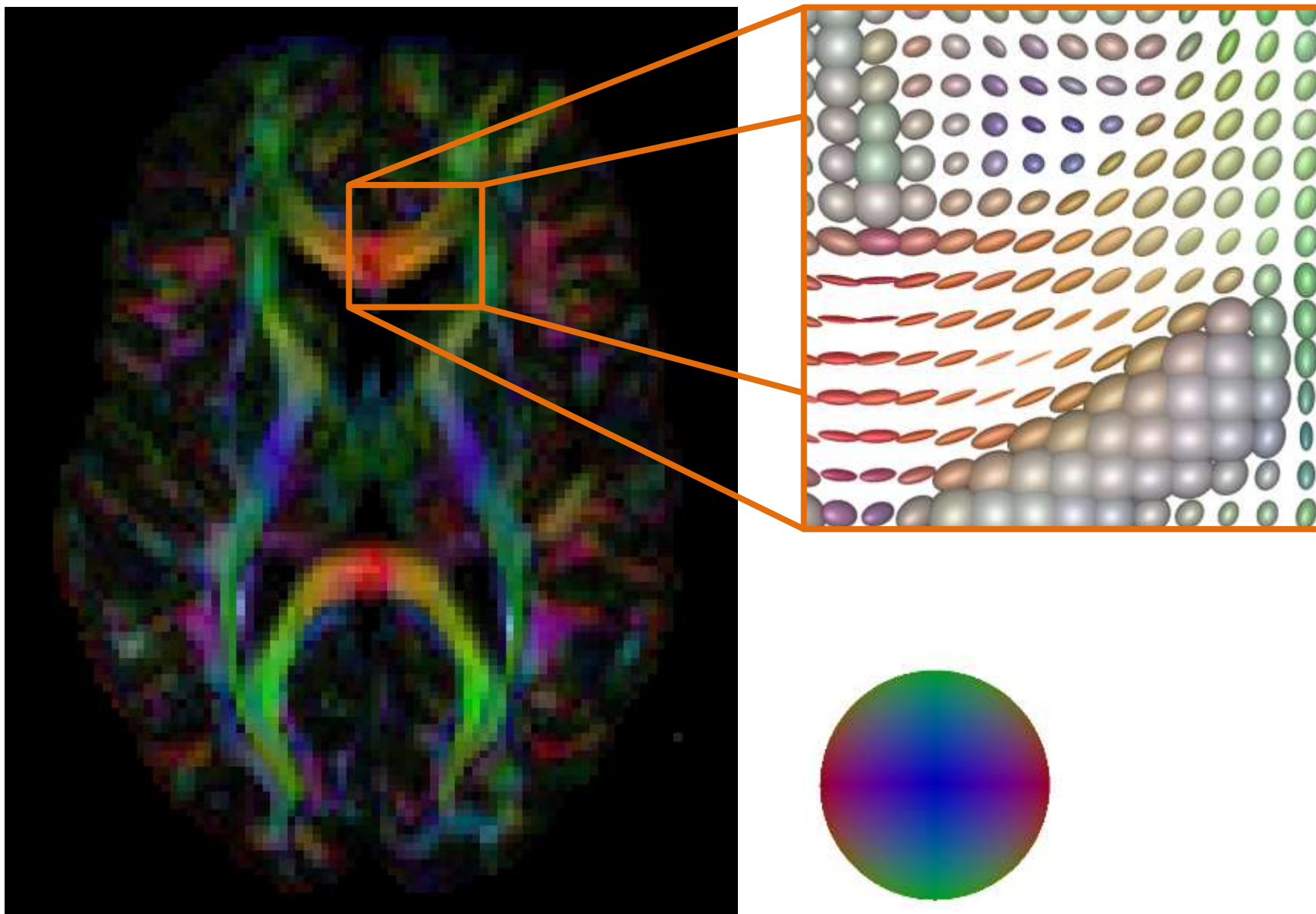
- **Problem:** In statistical analysis of FA maps, we would like to compensate incomplete alignment without smoothing
- **Tract-Based Spatial Statistics** is the standard approach to this problem
 - Restricts statistical analysis to a “skeleton”
 - Projection to that skeleton compensates for initial misalignment
 - Not perfectly, but better than other available methods

8.3 Fiber Tracking

Motivation: Fiber Tracking

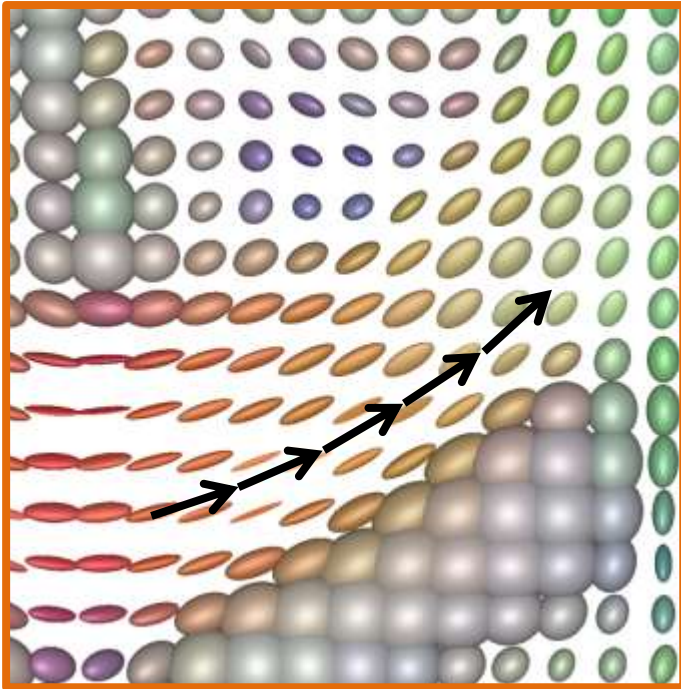
- **Fiber tracking / tractography** uses inferred local fiber directions to reconstruct the trajectories of major white matter tracts
 - Find out in which bundle a difference was localized
 - Perform statistical analysis along a bundle
 - Study fiber geometry itself (e.g., when planning surgery)



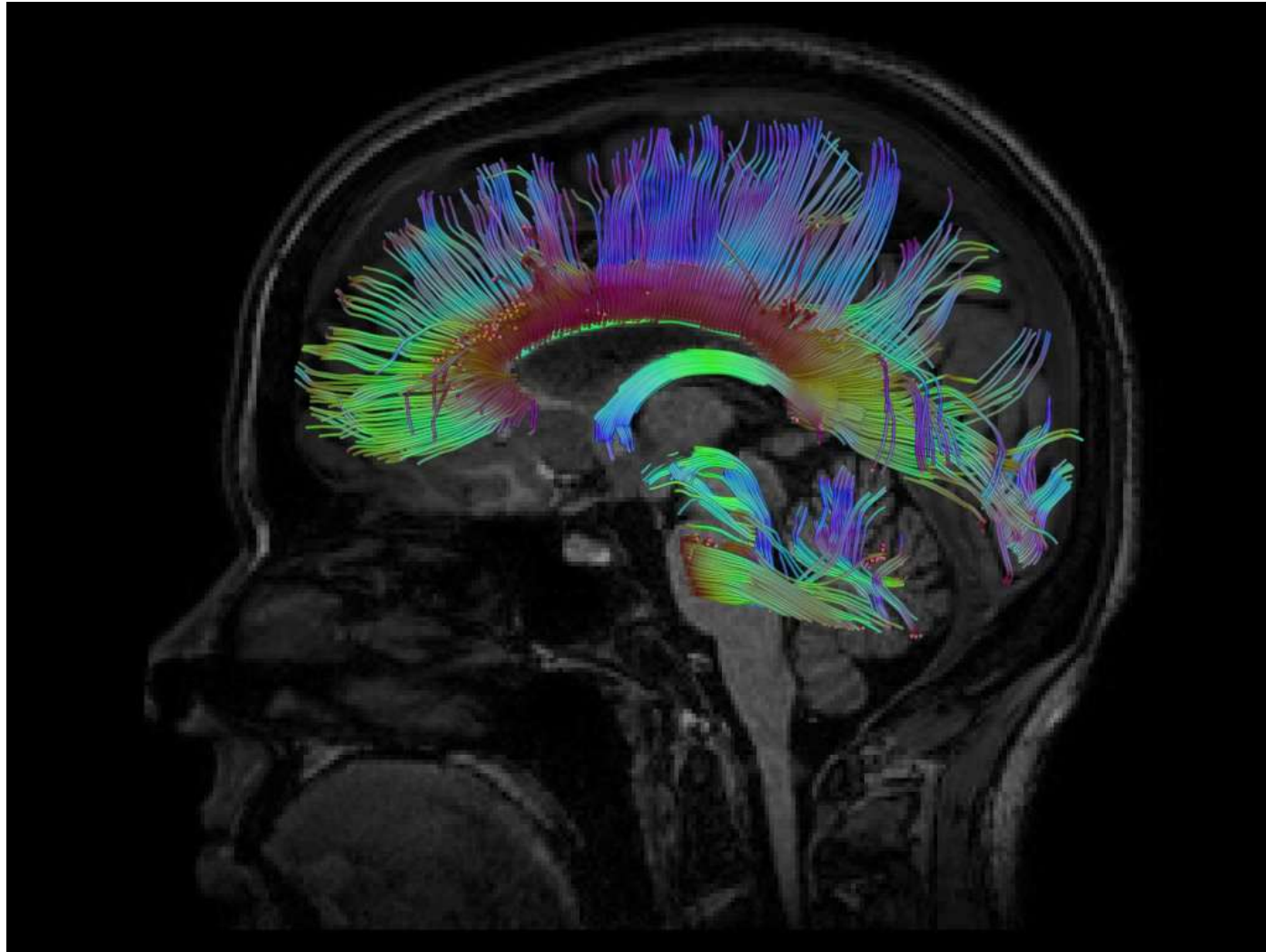


Deterministic Fiber Tracking

- In **deterministic tractography**, we proceed along a fixed tangent direction in each step



Sample Result: Deterministic Tracking



Ingredients in Fiber Tracking, Part 1

- A model for **local fiber orientation estimation**
 - *Example:* Principal DTI eigenvector, more sophisticated alternatives will be discussed later
- A numerical **integration scheme**
 - Deterministic Tractography can be viewed as solving the ordinary differential equation

$$\dot{\mathbf{x}}(t) = \mathbf{v}(\mathbf{x}(t))$$

- Euler integration with step size s :

$$\mathbf{x}_{i+1} = \mathbf{x}_i + s\mathbf{v}(\mathbf{x}_i)$$

- Account for the fact that diffusion MRI cannot estimate sign of fiber orientation $\mathbf{v}$
 - Always track forward, flip sign if needed
- Higher-order integration schemes (e.g., Runge-Kutta) also possible
 - Details left to Visual Data Analysis lecture (summer term)

Ingredients in Fiber Tracking, Part 2

- An **interpolation scheme** for fiber orientations
 - Integration evaluates $\mathbf{v}(\mathbf{x}(t))$ away from voxel centers
 - Nearest neighbor is simplest, but least accurate
 - Interpolating vectors needs to account for sign ambiguity to avoid cancellation
 - *Frequent alternative*: Interpolate diffusion tensor coefficients, re-compute eigenvectors
 - Or even interpolate measurements, re-fit the model

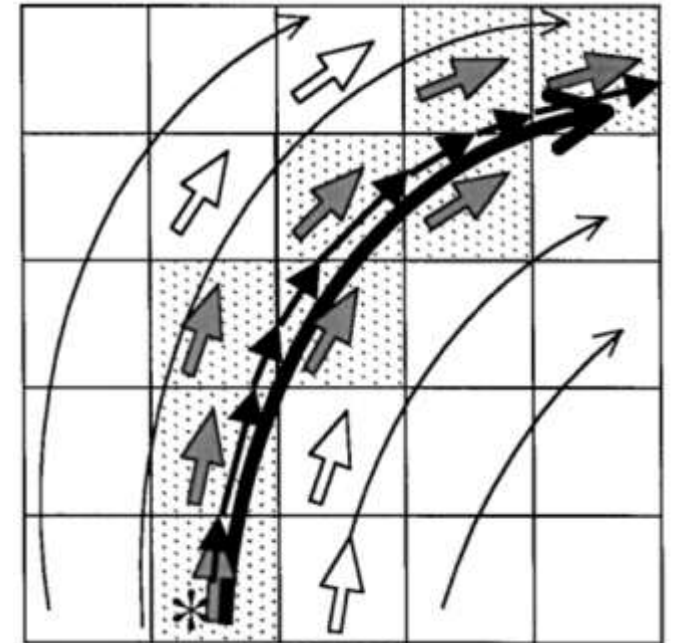
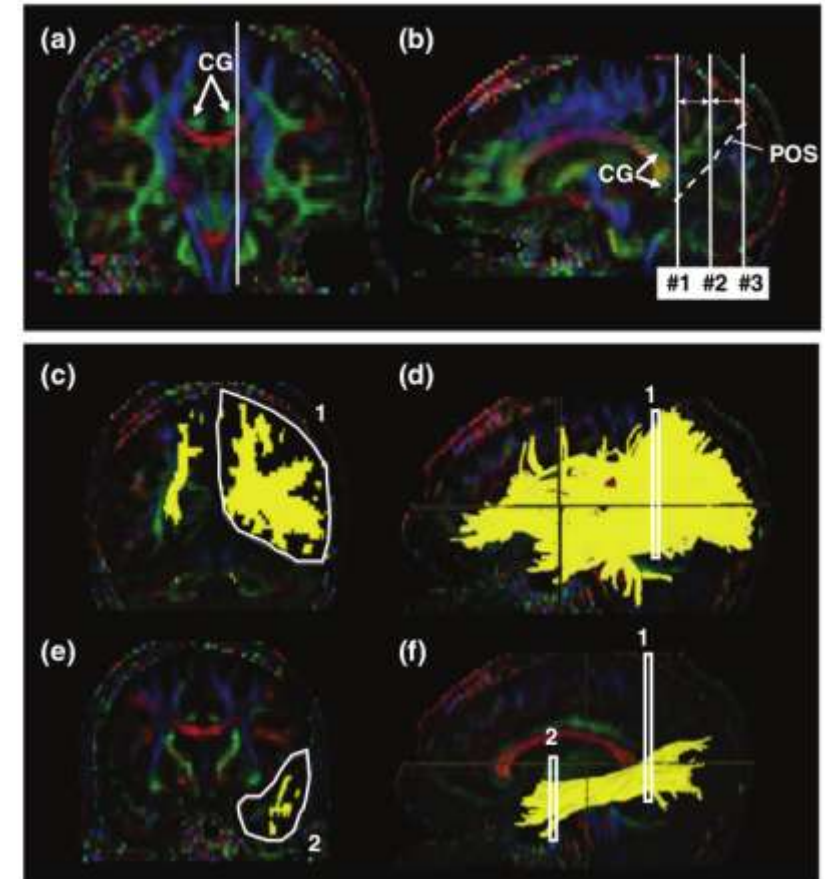


Image taken from Mori et al. [1999]

Ingredients in Fiber Tracking, Part 3

- **A seeding scheme**

- “Best practice” recommendations are available for major white matter tracts
- Define seeds in MNI space and transfer them via nonlinear registration
- *Alternative:* Seed everywhere and interactively select (“dissect”) or automatically classify afterwards

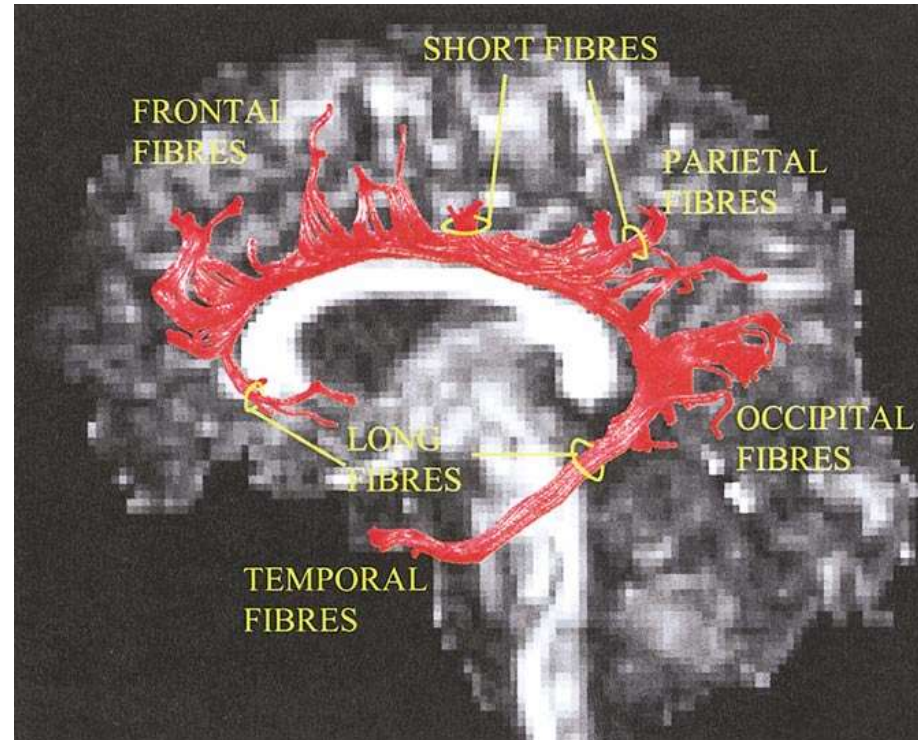


Ingredients in Fiber Tracking, Part 4

- Criteria for **track termination**
 - It is common to stop...
 - when anisotropy is low
 - when curvature is high
 - when leaving a white matter mask
- Criteria for **track acceptance**
 - It is common to discard...
 - very short tracks
 - tracks that terminate in white matter or CSF
 - tracks that do not conform with other anatomical priors

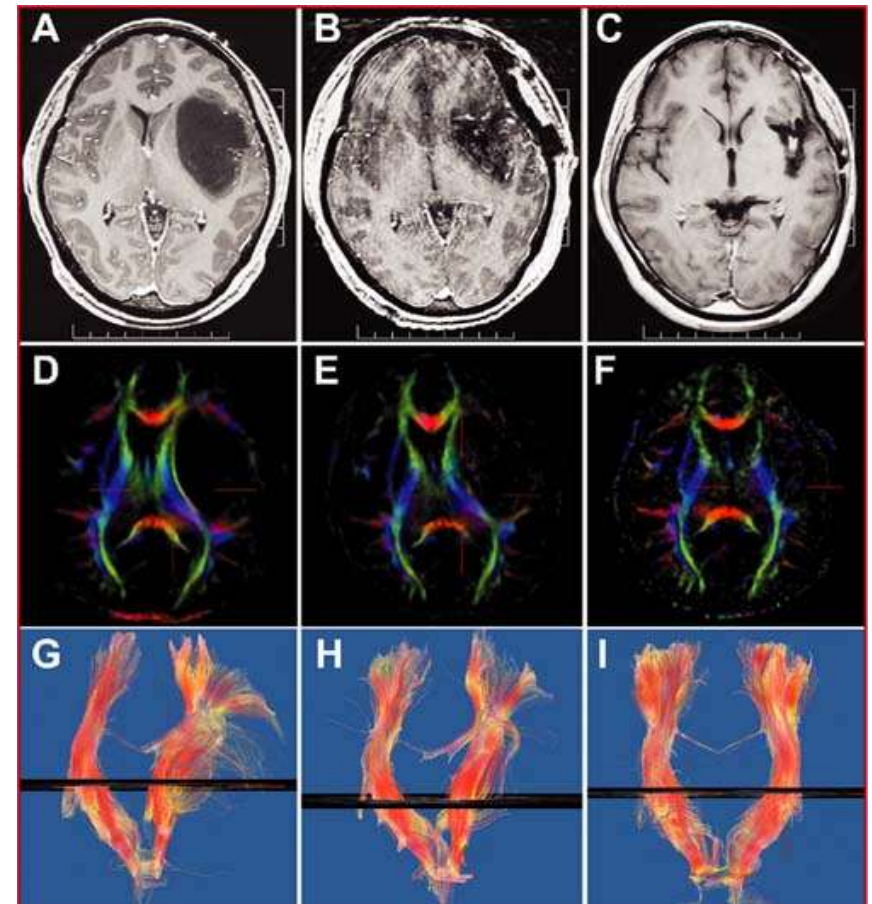
Success: White Matter Atlases

- **[Catani et al. 2002]** Deterministic tractography reliably reconstructs many previously known large-scale white matter tracts
 - Previously not possible to study them *in vivo*



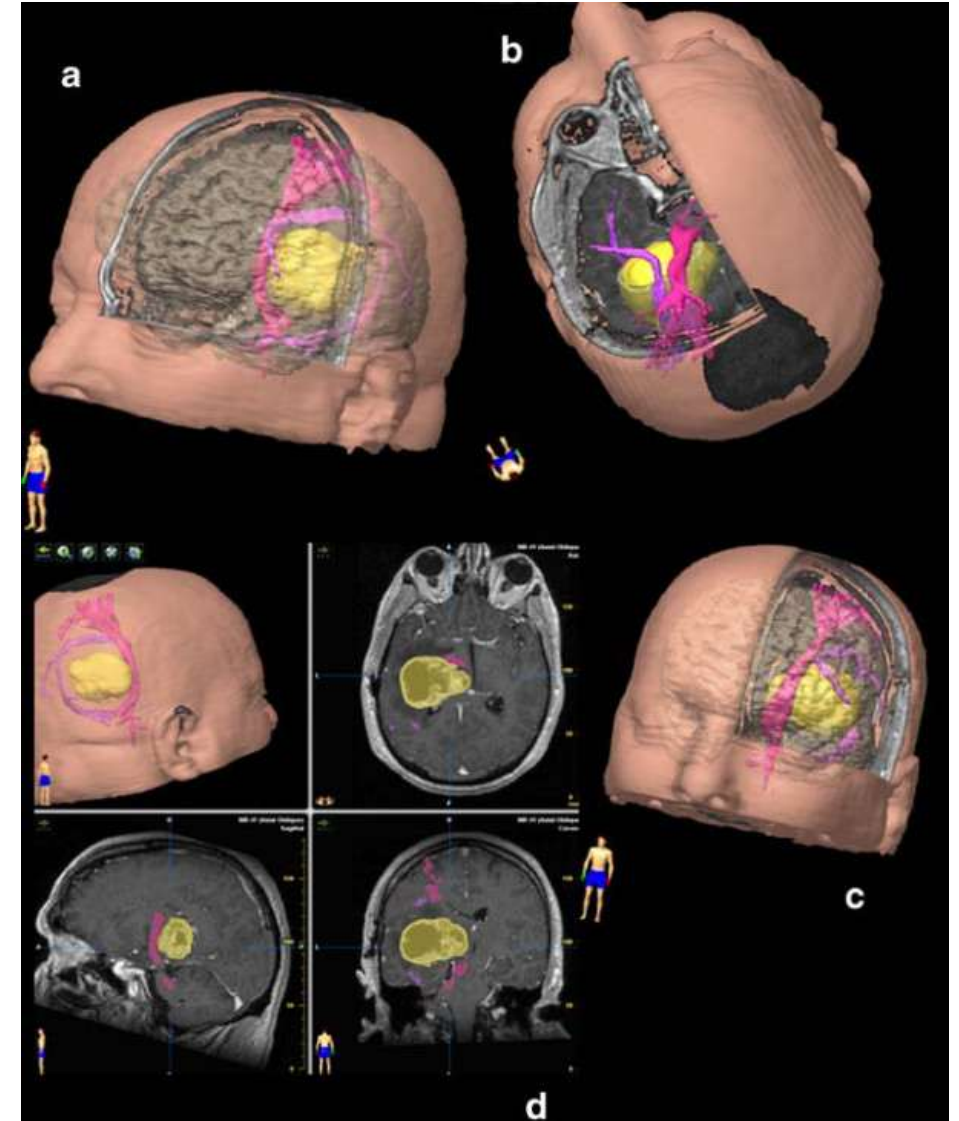
Success: Benefit in Surgical Planning

- [Wu et al. 2007] study the benefits of dMRI for planning **brain tumor surgery**. After randomly using (n=118) or not using (n=120) dMRI, it led to
 - Greater chance of complete resection
 - Reduced chance of postoperative motor deterioration
 - Longer survival
- **But:** Due to software constraints, mostly based on FA maps, fiber tracking only used in 23 cases



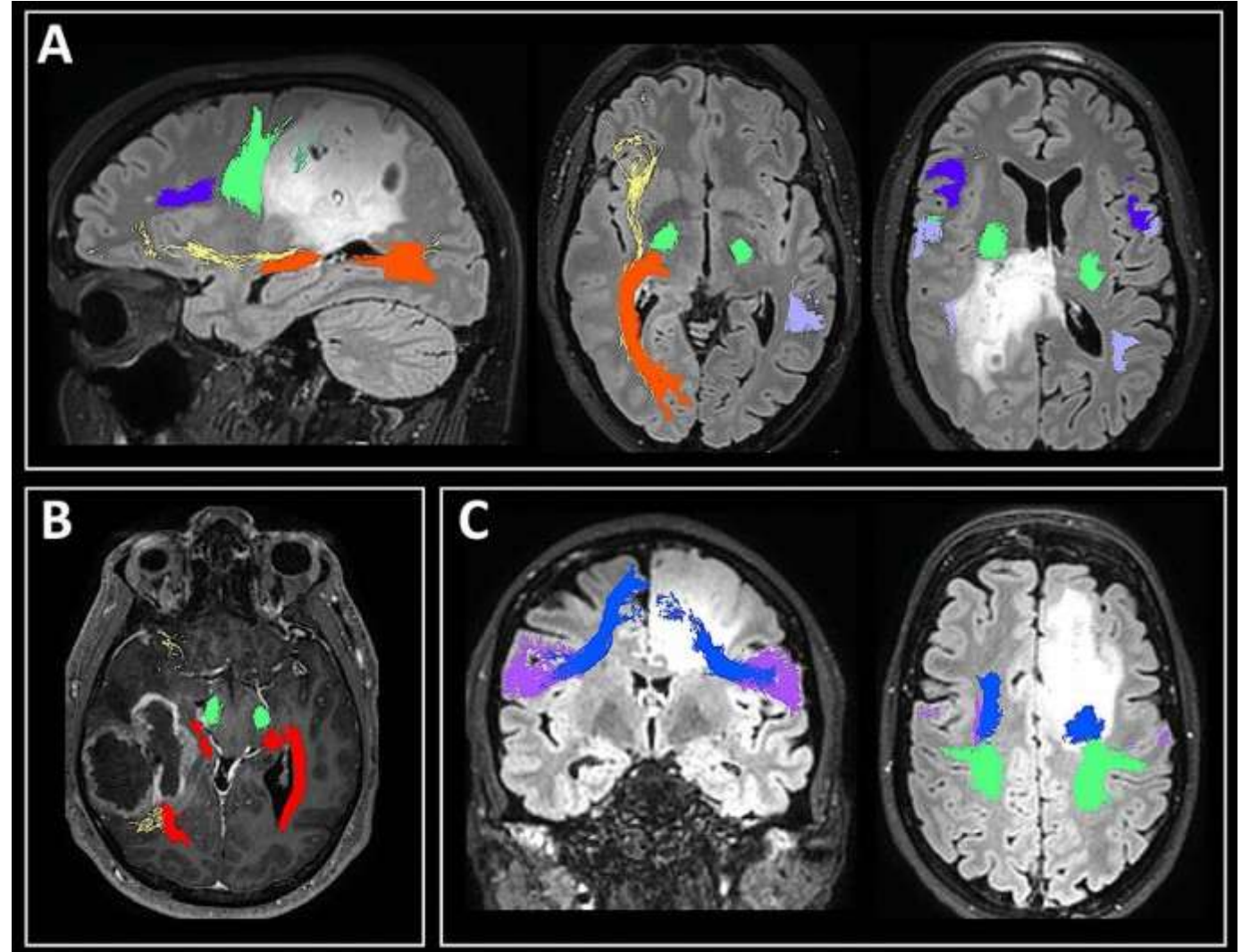
Success: Impact on Surgical Planning

- [Romano et al. 2009] report **impact percentages** of visualizing tractography in the vicinity of tumors while planning surgery on 37 patients
 - **Surgical approach** modified in 21%
 - **Resection margins** modified in 64%
 - **Overall impact: 82%**



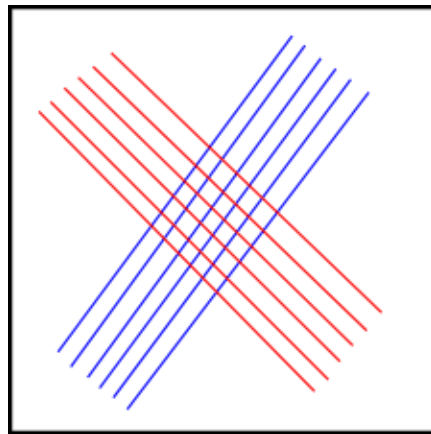
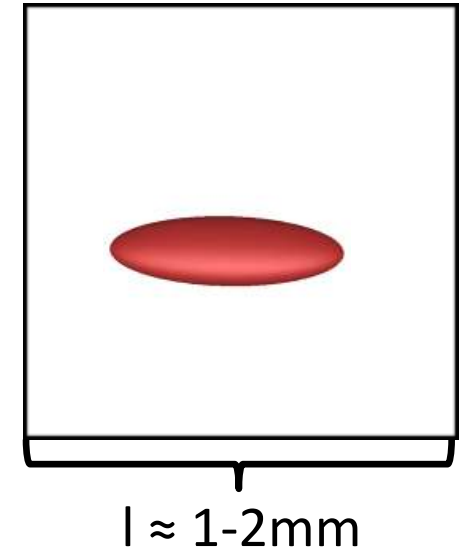
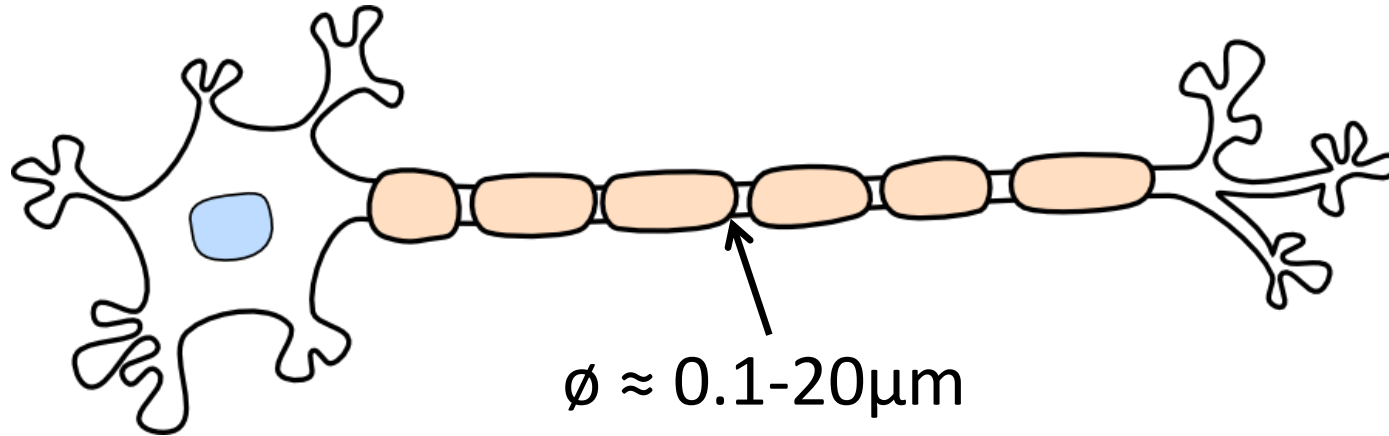
Challenge: Reliable Automated Workflow

- [Meesters et al. 2023] observe under-exploitation of tractography in clinical practice due to
 - lack of automation, standardization
 - need for specialized software and expertise

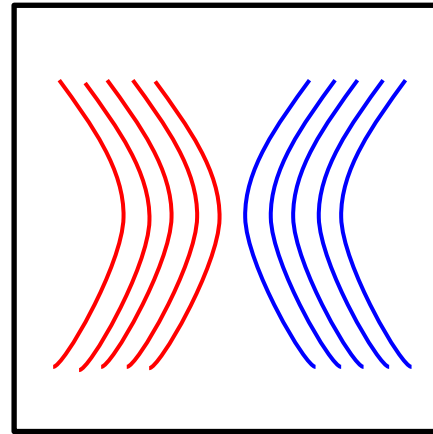


8.4 Multi-Fiber Tracking

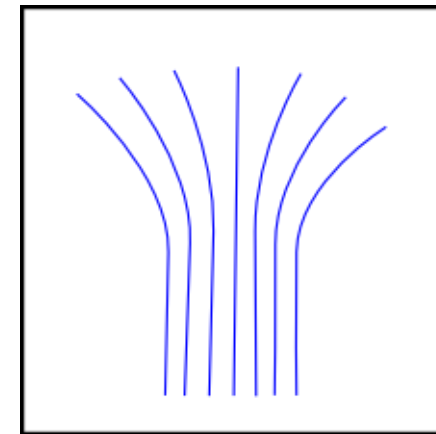
Partial Voluming: More than One Fiber per Voxel!



Crossing Fibers



"Kissing" Fibers



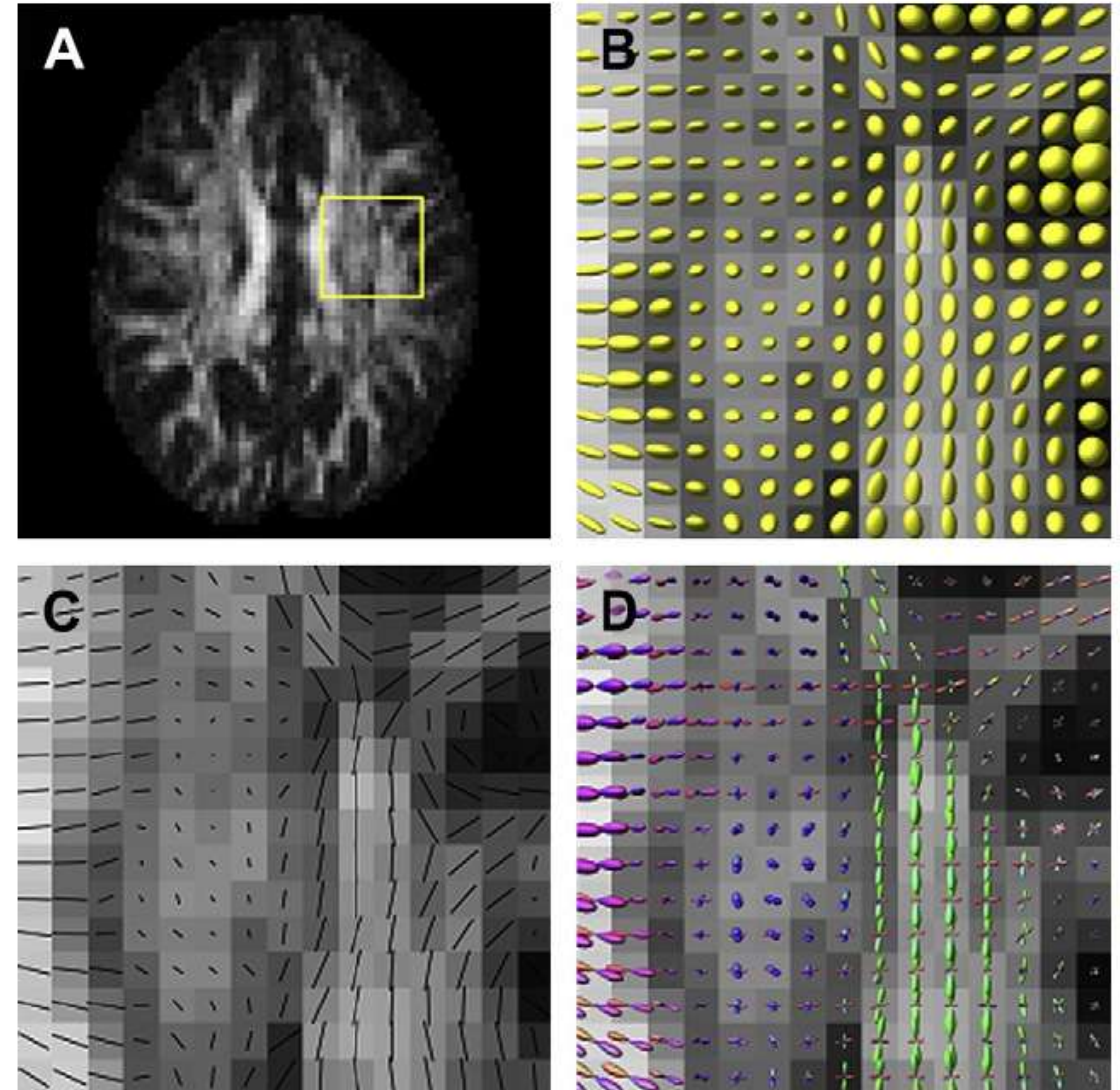
Diverging Fibers

How Common are Crossing Fiber Voxels?

- **No unique answer:** How many compartments can be reliably distinguished depends on many factors, including
 - Measurement (b value, SNR, spatial resolution)
 - How we estimate the number of compartments
- All authors agree that the fraction of multiple fiber voxels is substantial, and that **fiber tracking should account for it**
 - [Behrens et al. 2007] 1:~70% 2:~30%
 - [Jeurissen et al. 2010] 1:~10% 2:~40% 3:~50%
 - [Schultz 2012] 1:~16% 2:~44% 3:~40%

Multi-Fiber Tracking with DTI?

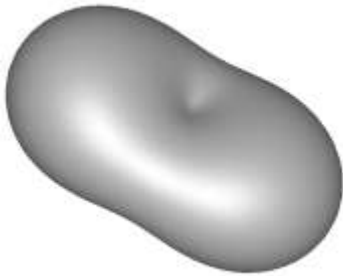
- “Fat” diffusion tensors indicate that there may be a crossing, but do not indicate its direction(s)



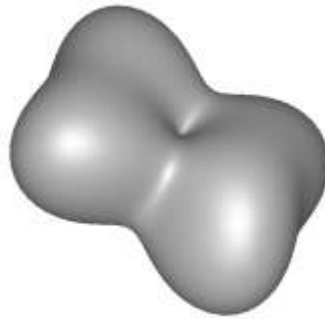
Higher-Order ADC Models

Initial idea: Model $D(\mathbf{x})$ as a Higher-Degree Polynomial

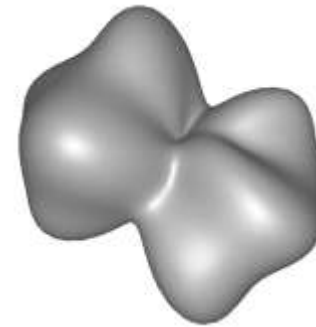
$$S(D(\mathbf{x})) = S_0 e^{-bD(\mathbf{x})}$$



Degree 2



Degree 4

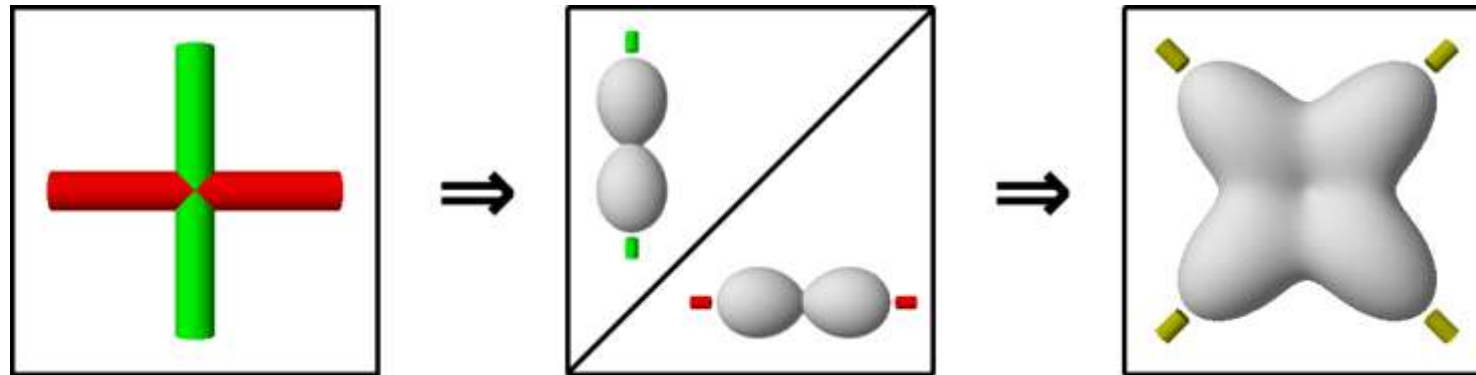


Degree 6

Note: Due to limitations in diffusion MRI protocols, all models will assume antipodal symmetry: $D(\mathbf{x}) = D(-\mathbf{x})$

An Important Caveat

When a voxel contains more than one fiber compartment, ADC maxima **no longer** align with fibers!



$$S(\mathbf{x}) = S_0 \sum_{i=1}^2 f_i e^{-b\mathbf{x}^T \mathbf{D}_i \mathbf{x}}$$

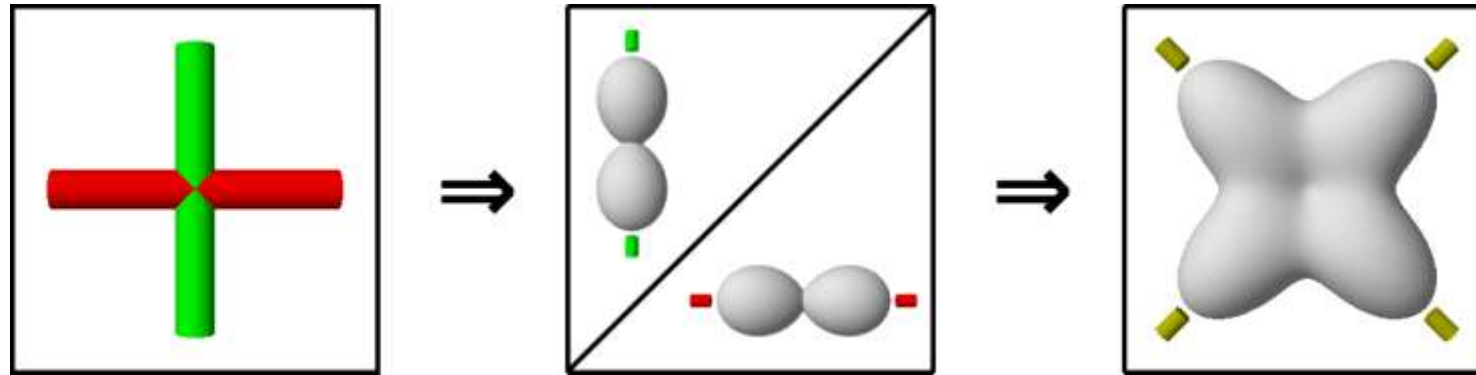
$$S(\mathbf{x}) = S_0 e^{-bD(\mathbf{x})}$$

Overview: Multi-Fiber Models

We will discuss **two families of multi-fiber models**:

- 1. Multi-Tensor Models**
2. Models based on deconvolution

Motivate Multi-Tensor Models



$$S(\mathbf{x}) = S_0 \sum_{i=1}^2 f_i e^{-b \mathbf{x}^T \mathbf{D}_i \mathbf{x}}$$

$$S(\mathbf{x}) = S_0 e^{-b D(\mathbf{x})}$$

Acquisition Strategies: DTI vs. HARDI

- **Diffusion Tensor Imaging (DTI) Acquisition**

- $b_{\text{low}}=0\text{mm}^2/\text{s}$, $b_{\text{high}}\in[750,1300]\text{mm}^2/\text{s}$
- At least 6 non-collinear directions at b_{high}
 - A larger number (≈ 30) is often used to reduce variance
- Number of acquisitions at $b_{\text{high}} : b_{\text{low}} \approx 6:1$

- **High Angular Resolution Diffusion Imaging (HARDI)**

- Aim for higher angular resolution than DTI, acquire 60-100 directions
- Acquiring a single high b value sufficient to capture angular contrast
- Larger b values provide better angular contrast,
 $b_{\text{high}}\in[1000,3000]\text{mm}^2/\text{s}$

The Ball-and-Stick Model

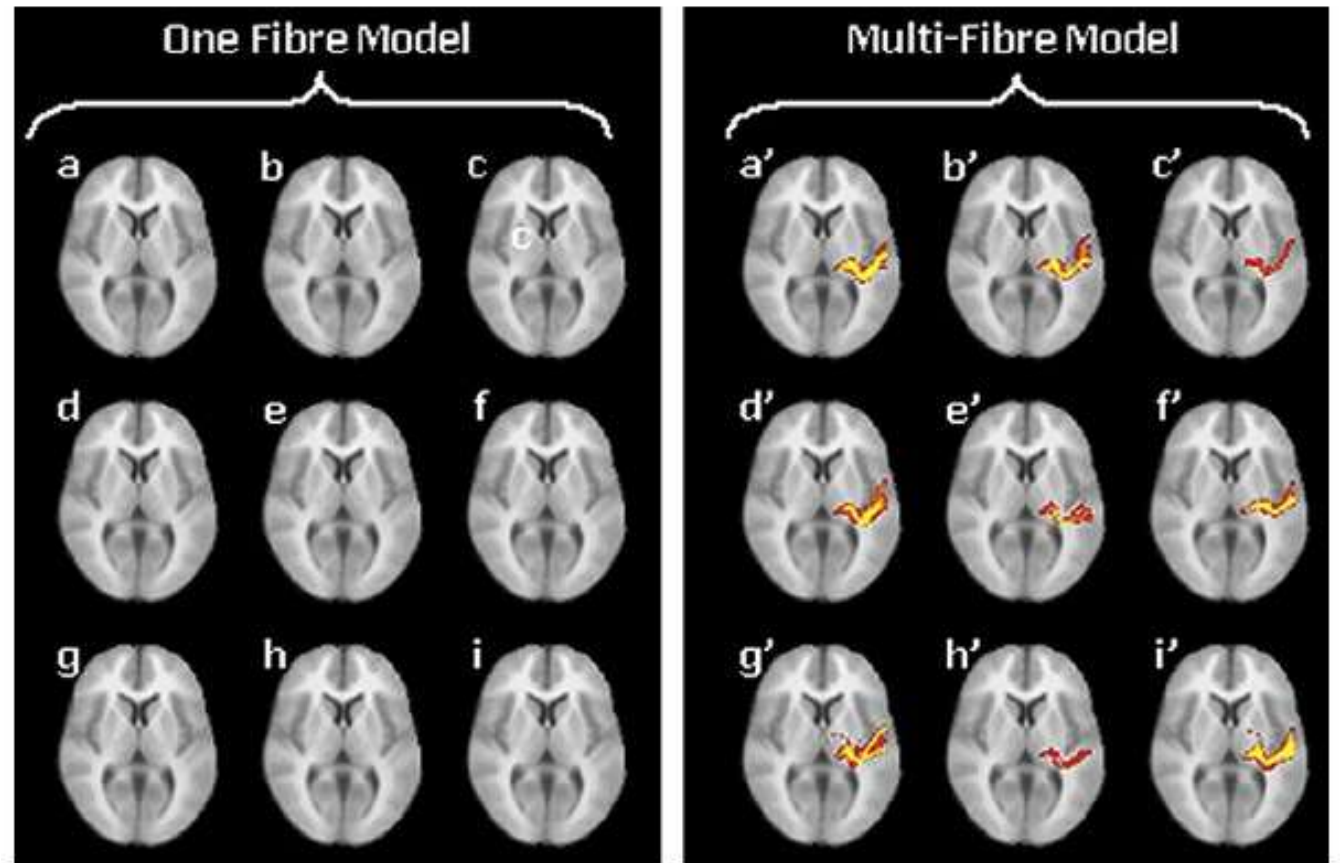
$$S(\mathbf{x}) = S_0 \sum_{i=1}^k f_i e^{-b\mathbf{x}^T \mathbf{D}_i \mathbf{x}}$$

Since it is impossible to simultaneously determine the volume fractions and isotropic parts of different fiber compartments, Behrens et al. constrain multi-tensor model to a **ball and** one [2003] or multiple [2007] **sticks**:

- “Ball” is perfectly isotropic ($\mathbf{D}=d\mathbf{I}$)
- “Sticks” are perfectly linear ($\mathbf{D}=d\mathbf{v}\mathbf{v}^T$)
 - Diffusivity d is assumed to be the same in all compartments
- Fitting becomes non-linear (e.g., Levenberg-Marquardt, Ch.4)
- Behrens et al. use this in a Bayesian fiber tracking framework

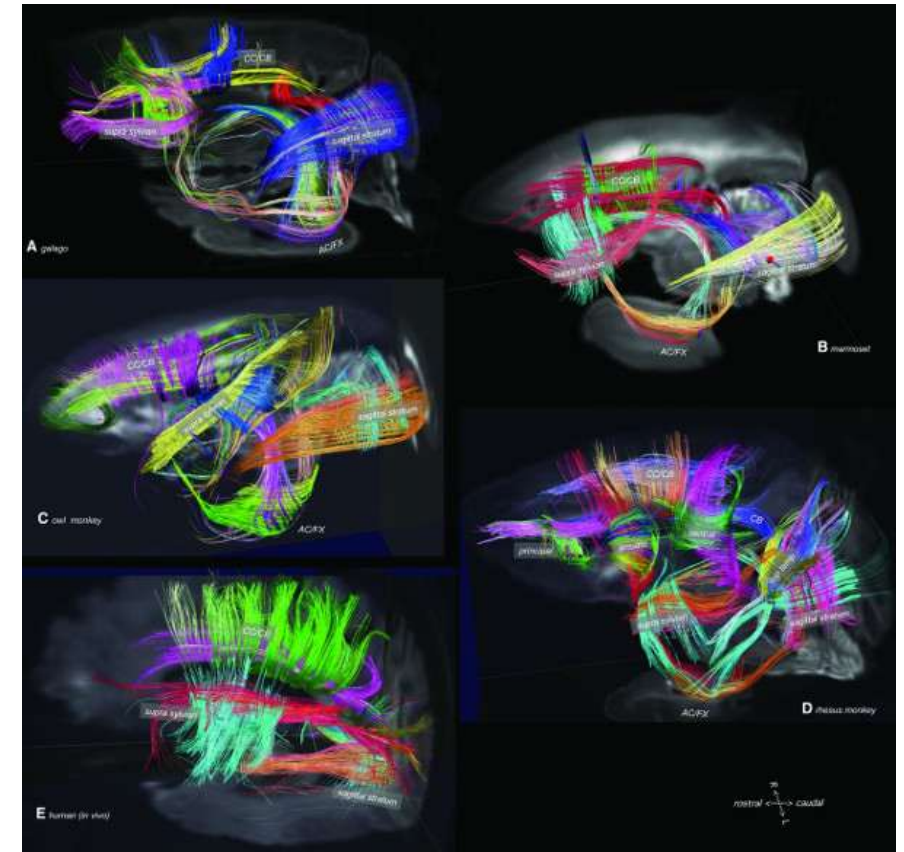
Multi-Fiber Tractography

- When **multiple fiber estimates** are available:
 - Follow the one most collinear with the incoming direction
 - When in doubt, consider branching
 - Consider regularizing noisy multi-fiber estimates
- Some bundles (such as acoustic radiation) can only be found using multi-fiber models



Crossing Fibers and White Matter Architecture

- **[Wedeen et al. 2012]** used fiber tracking for a comparative study of white matter geometry in humans and several non-human primates
 - 3D grid of parallel sheets of near-orthogonally interwoven paths
 - Strong cross-species homologies
 - Postulate three principal axes of development



Pros and Cons of Multi-Tensor Models

Advantages of Multi-Compartment Models:

- Natural generalization of diffusion tensor model
- Directly provides estimates of fiber directions

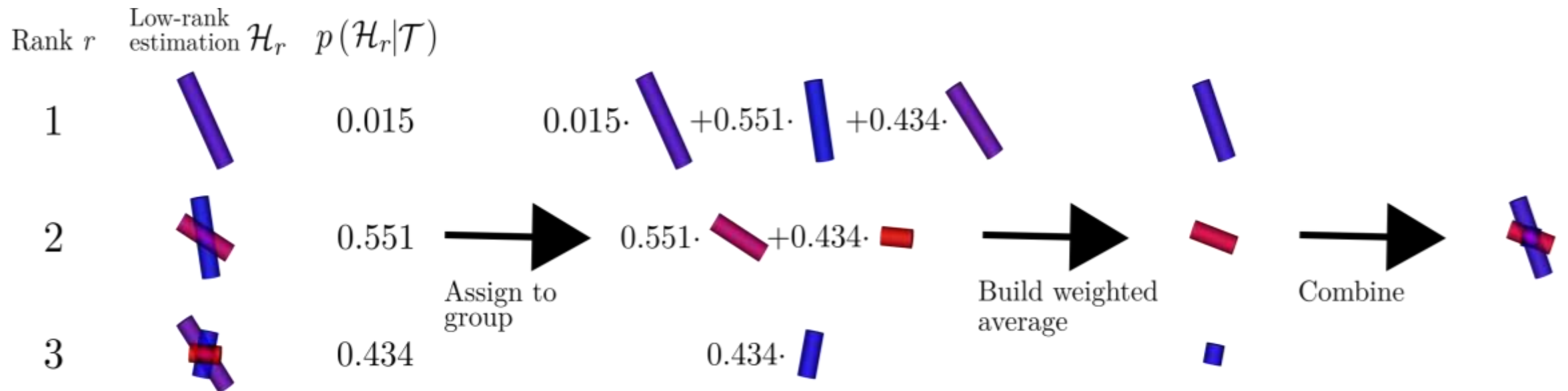
Disadvantages of Multi-Compartment Models:

- Model assumptions are often rather restrictive
 - e.g., same diffusivity in all compartments
- Require nonlinear fitting (less reliable and efficient)
- Need to select the appropriate number of fibers

Reducing Model Uncertainty

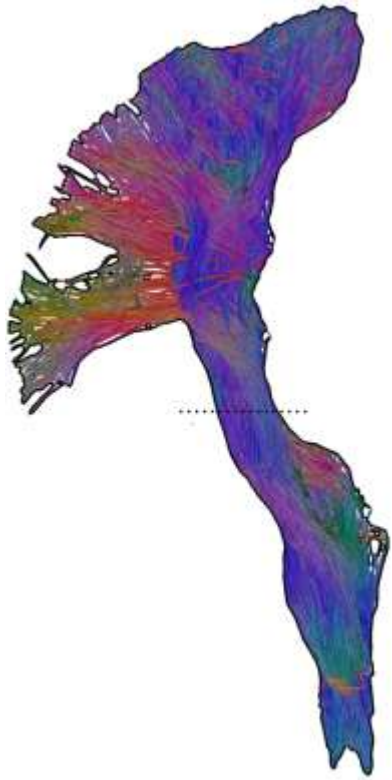
[Grün et al. 2023] replace the selection of a fixed value of k in each integration step with a **weighted average** of tracking directions from different values of k

- Weights are posterior probabilities of k given the local data

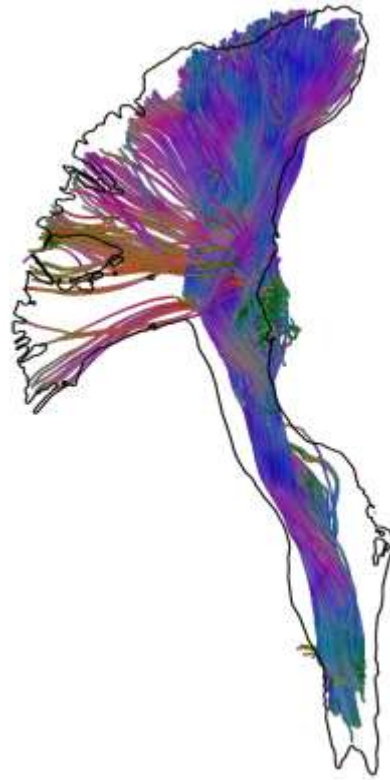


Reducing Model Uncertainty: Qualitative Results

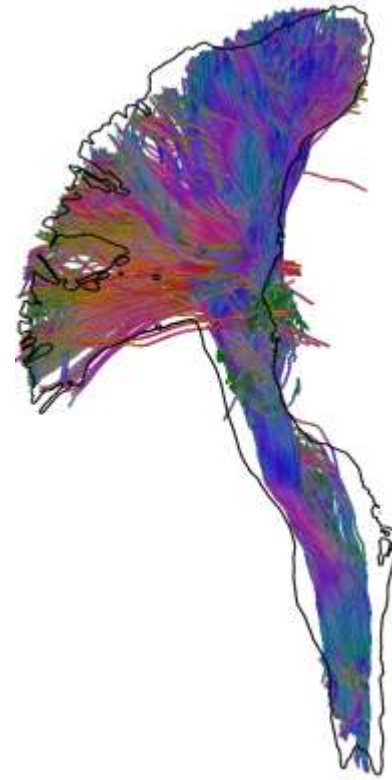
Visual comparison of reconstructing the right corticospinal tract from seeds in the plane indicated by the dashed line:



Curated Reference



Model Selection



Model Averaging

Overview: Multi-Fiber Models

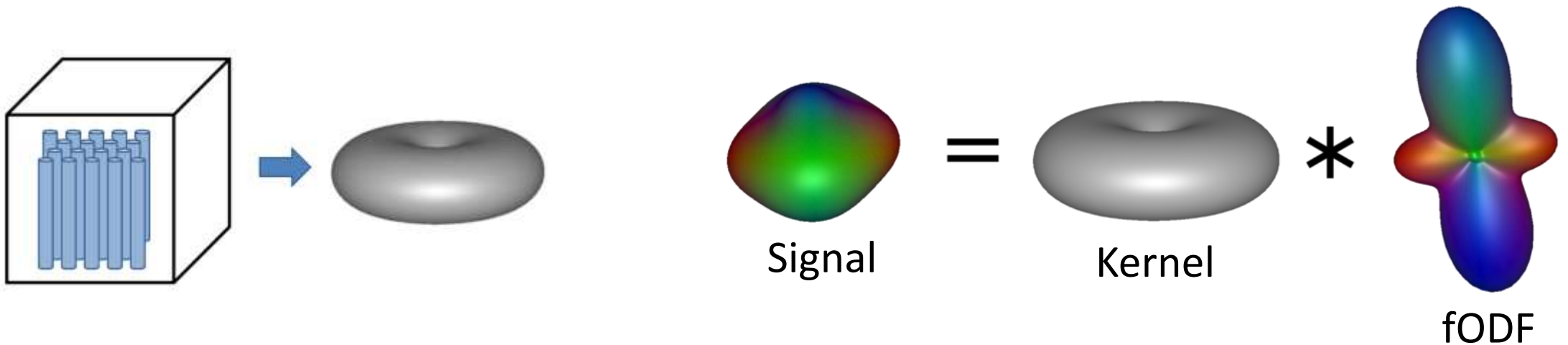
We will discuss **two families of multi-fiber models**:

1. Multi-Tensor Models
2. **Models based on deconvolution**

Spherical Deconvolution: Basic Idea

Observation by Behrens et al. [2003]:

- If, up to their orientation, all fibers contribute the same MR signal, and there is no exchange between them, the overall signal is the **spherical convolution** of a fiber orientation distribution function (fODF) with a kernel that reflects the MR signal from a *single fiber compartment*



Spherical Convolution: Formalization

- **Intuition** of spherical convolution between the fiber ODF $F(\theta, \phi)$ and an axially symmetric kernel $R(\theta)$:
 - Rotate a copy of R to align its axis with any direction (θ', ϕ') on the sphere
 - Scale its contribution by the local value of $F(\theta', \phi')$
 - Integrate the contributions of all rotated and scaled copies
- The resulting **convolution integral** can be written as

$$S(\theta, \phi) = \int_0^{2\pi} \int_0^\pi F(\theta', \phi') R(\gamma') \sin(\theta') d\theta' d\phi'$$

- γ' denotes the angle between the directions given by (θ, ϕ) and (θ', ϕ')

Reminder: $f(x) = (h * g)(x) = \int_{-\infty}^{\infty} h(\xi) \cdot g(x - \xi) d\xi$

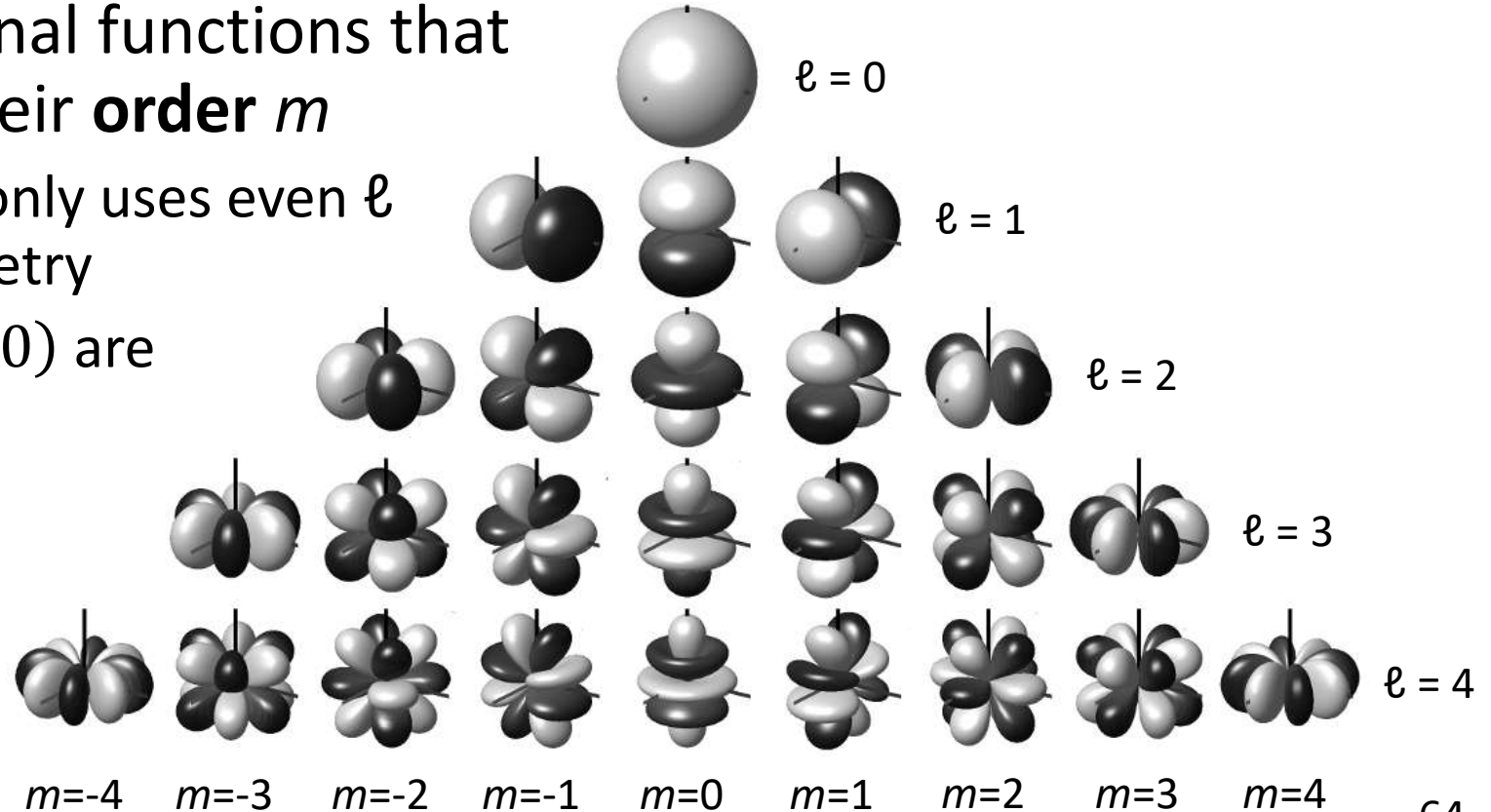
Spherical Harmonics

Spherical harmonics (SH) provide a set of **orthonormal basis functions** on the sphere that is analogous to the Fourier basis

- **Degree ℓ** indicates angular frequency band, contains $2\ell+1$ orthogonal functions that are distinguished by their **order m**

- Diffusion MRI analysis only uses even ℓ due to antipodal symmetry
- Zonal harmonics ($m = 0$) are axially symmetric

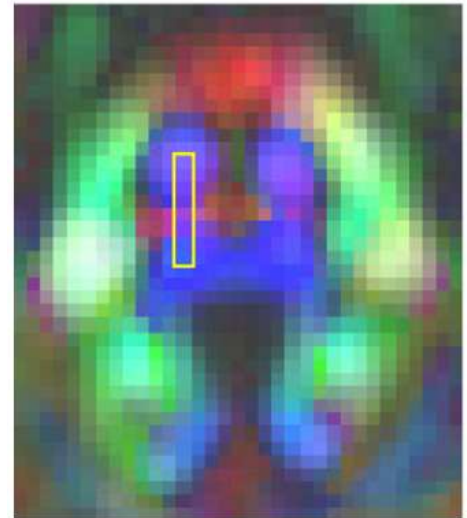
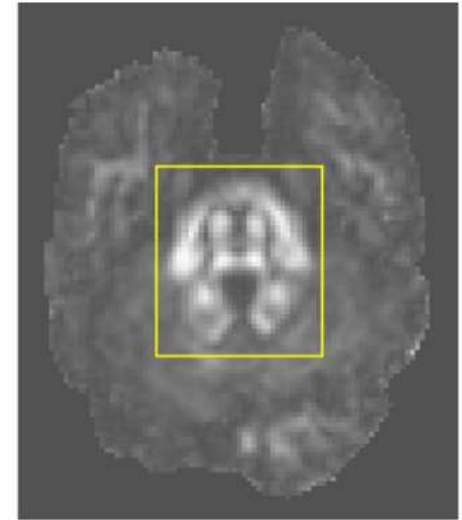
- **Convolution theorem** facilitates spherical (de)convolution



Spherical Deconvolution: Implementation

Tournier et al. [2004] used the following pipeline to estimate multiple fiber compartments:

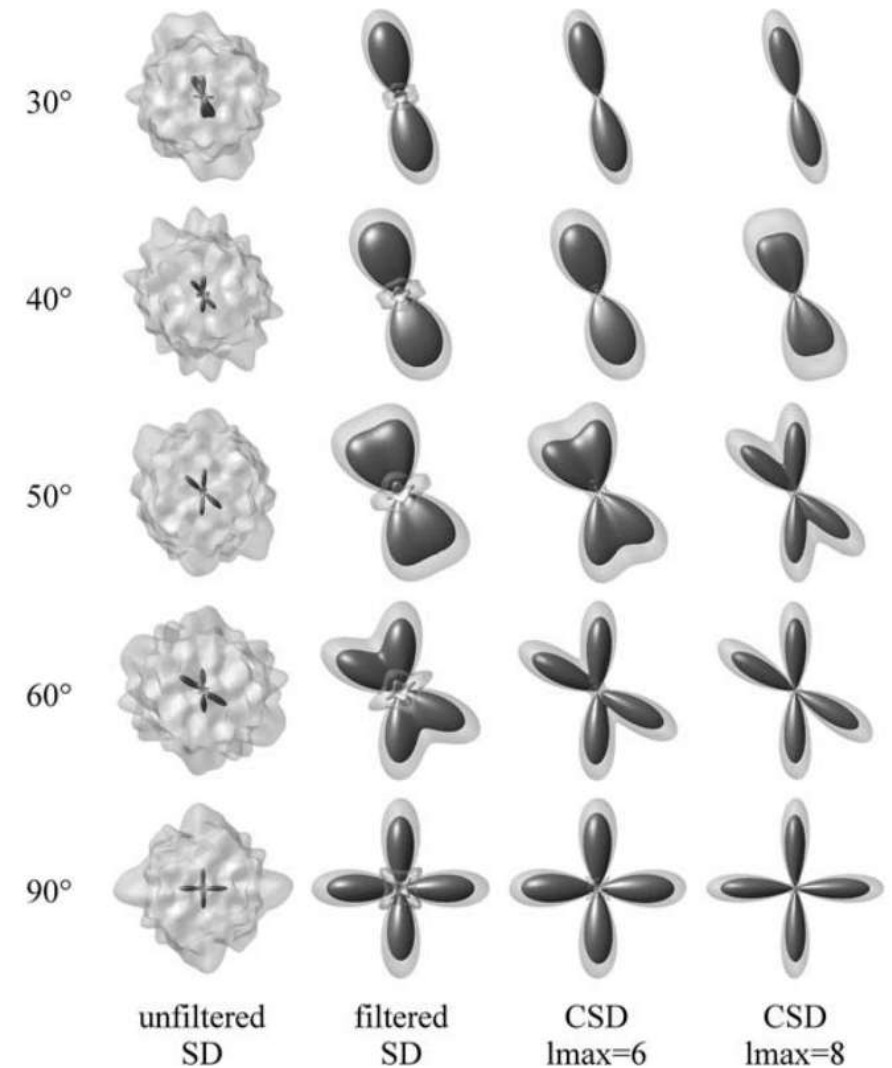
1. Linear least-squares fit of spherical harmonics to the HARDI measurements in each voxel
2. Estimate single-fiber response from putative single-fiber voxels (high FA in DTI analysis)
3. Compute fODFs via SH convolution theorem
 - *Main challenge:* Division by small values for large ℓ greatly amplifies measurement noise. Solved by low-pass filtering the result
4. fODF maxima are taken as fiber directions, their magnitude as volume fractions



Constrained Spherical Deconvolution: Idea

Tournier et al. [2007] demonstrates that a non-negativity constraint permits omitting the low-pass filter on fODFs

- Negative fODF values clearly make no sense
- In practice, even small positive values are penalized. This is motivated by the assumption of fODF sparsity
- Can increase angular resolution by including higher SH degrees



Constrained Spherical Deconvolution: Implementation

In each voxel:

1. Perform filtered deconvolution to obtain an initial fODF $\mathbf{f}_0$
2. Create a set N_0 of directions with negative fODF values based on checking 300 uniformly distributed directions

3. Iterate until convergence:

- a. Solve the following regularized least squares problem:

$$\mathbf{f}_{i+1} = \operatorname{argmin}_{\mathbf{f}} \|\mathbf{A}\mathbf{f} - \mathbf{b}\|^2 + \lambda^2 \|\mathbf{L}_i\mathbf{f}\|^2$$

- $\mathbf{A}$ convolves fODF $\mathbf{f}$ with the single-fiber kernel and evaluates the result in all measurement directions, $\mathbf{b}$ contains the diffusion-weighted measurements
- $\mathbf{L}_i$ evaluates $\mathbf{f}$ in all directions contained in N_i , λ is a regularization weight

- b. Update N_{i+1} based on $\mathbf{f}_{i+1}$

Pros and Cons of Fiber ODF Models

Advantages of Fiber ODF Models:

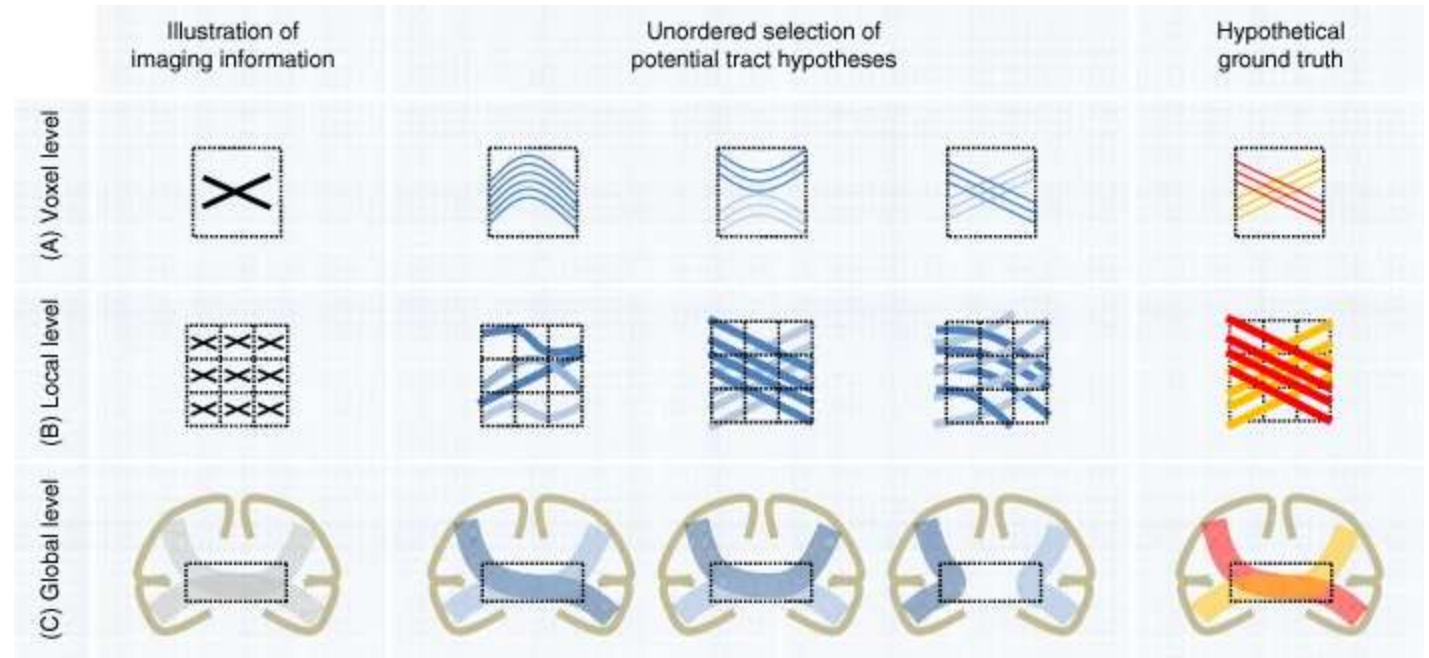
- Do not require pre-specifying the number of fiber compartments in a given voxel
- Sharpness of fODF peaks models spread in addition to crossings

Disadvantages of Fiber ODF Models:

- Strong model assumptions
 - Fiber response often assumed to be uniform over the whole brain
- Deconvolution is a numerically ill-conditioned operation that requires constraints or regularization
- Angular resolution more limited than in multi-tensor models

Remaining Challenges of Multi-Fiber Tractography

- **Validation** remains an important challenge in tractography
- Based on experiments on simulated human brain data with known ground truth [Maier-Hein et al. 2017]
 - Most multi-fiber tractography methods recover around 90% of the true bundles (to some extent)
 - Most whole-brain results contain more invalid than valid bundles
 - Many invalid bundles occur systematically
 - **Reason:** Local imaging information insufficient for unambiguous tracking



Summary: Multi-Fiber Tractography

State-of-the-art **fiber tractography** accounts for multiple fiber compartments per voxel. Popular multi-fiber models include:

1. Multi-Tensor Models

- + Modest demands on acquisition, immediately yields fiber directions
- Nonlinear fitting, choice of k , strong assumptions in ball-and-stick

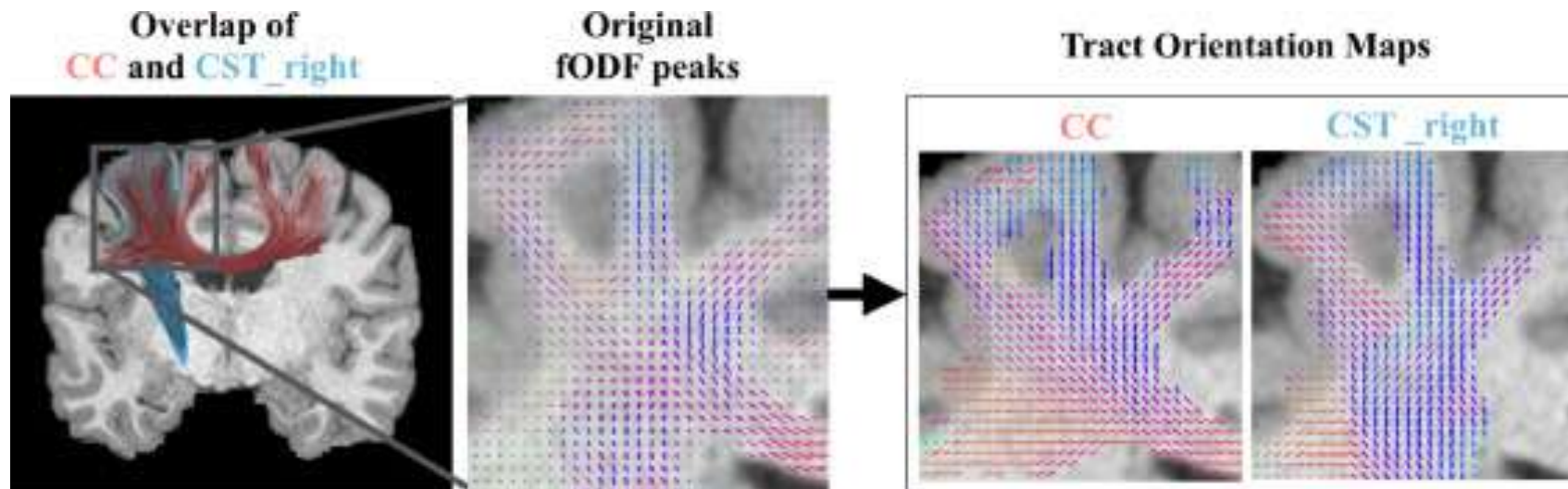
2. Models based on deconvolution

- + Simultaneously estimates number and direction of fiber compartments, accounts for spread
- Poor conditioning, limited angular resolution

8.5 Machine Learning based Tractography

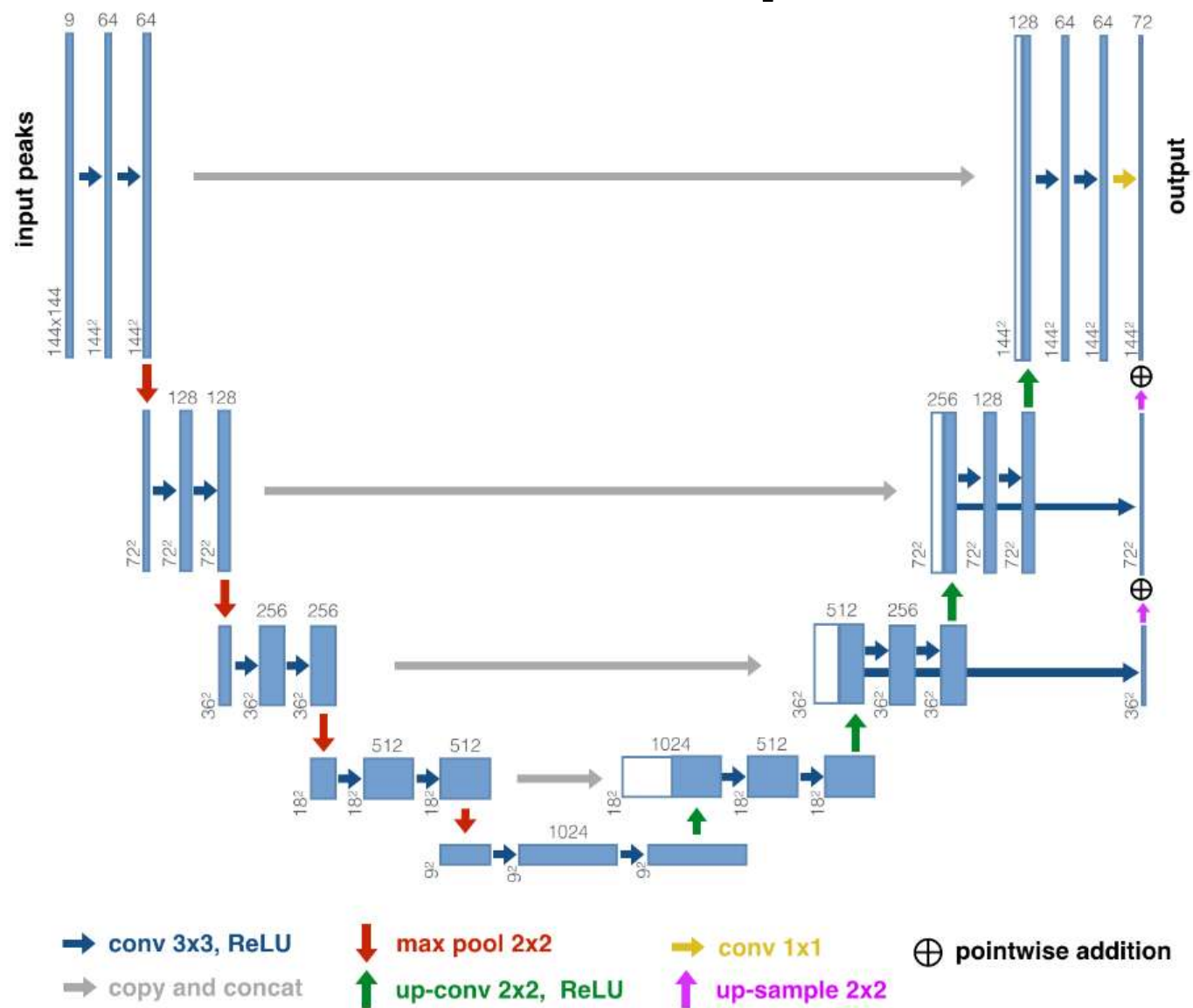
Idea: Tract Orientation Maps

- A key ambiguity in multi-fiber tractography arises from having multiple potential tracking directions to choose from
- **Tract Orientation Maps** are bundle-specific, contain at most one direction per voxel
 - Permits automated *bundle-specific* tractography
 - Assignment requires anatomical knowledge, permits regularization



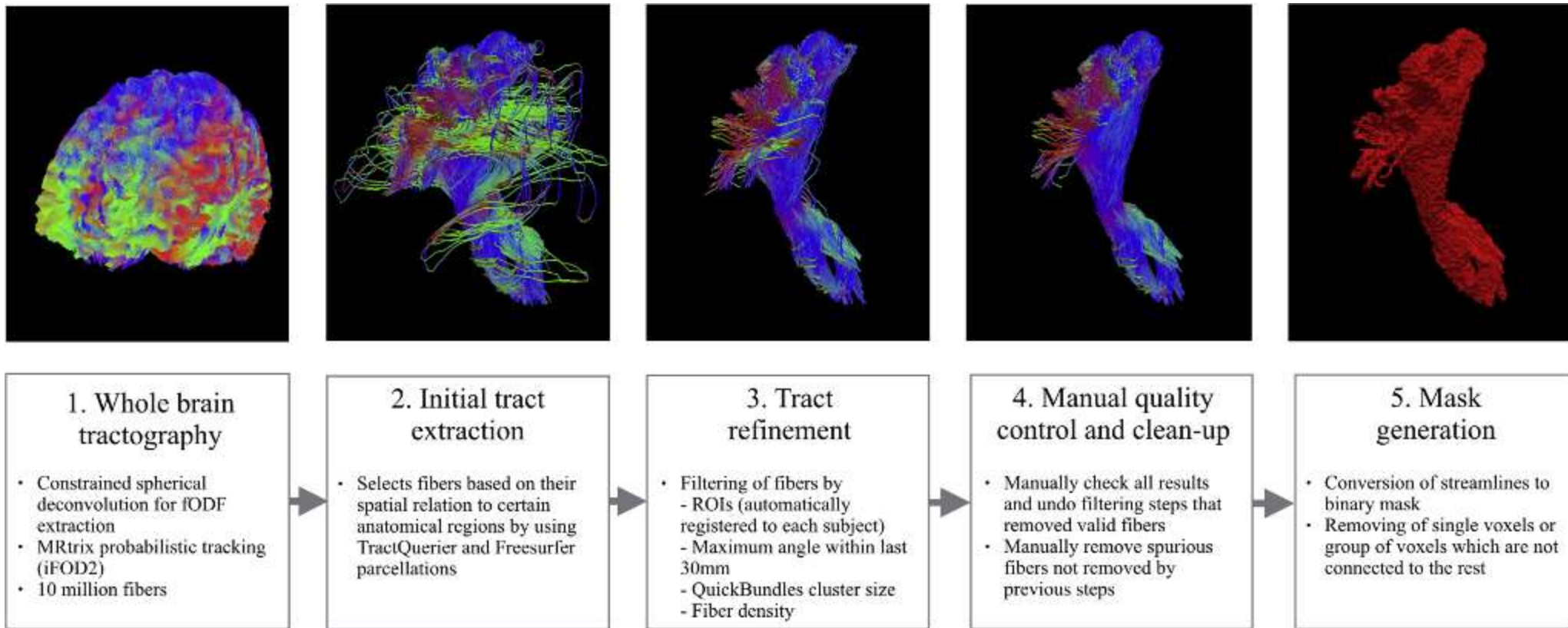
Learning to Create Tract Orientation Maps

- [Wasserthal et al., MedIA 2018] train a **U-Net-like architecture** to map three main fiber directions from constrained spherical deconvolution to 72 Tract Orientation Maps
 - Based on coronal slices
 - Train four networks for 18 tracts each
- Differences from U-Net:
 - Deep supervision
 - Padded convolutions



Training Data

- As a basis for training and evaluation, 72 tracts each have been semi-automatically dissected in 105 subjects
 - TOM reference direction = mean of largest cluster in that voxel



Tractography vs. Segmentation

- **Problem:** TOM alone tends to make bundles too wide
- **Remedy:** Additionally segment volumetric extent of each tract with a separate CNN, enforce that mask during fiber tracking
 - **TractSeg** [Wasserthal et al., NeuroImage 2018]
- **Observation:** In many use cases, all relevant information is even retained in binary segmentation masks
 - In particular, this is true for the two main motivations we gave:
 1. Using bundles as a basis for statistical analysis (e.g., of FA)
 2. Knowing the spatial extent of a bundle (e.g., for surgery)

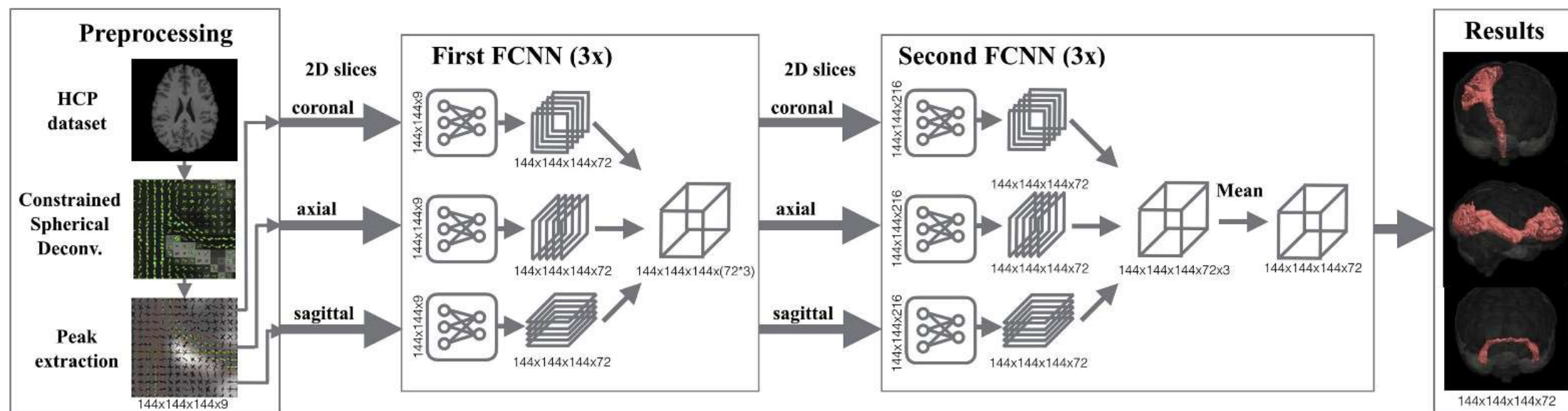
TractSeg Architecture

To adapt the U-Net to diffusion MRI (dMRI), TractSeg makes the following changes:

- **Input:** Reduces the large number of dMRI measurements to the three main peaks per voxel via constrained spherical deconvolution
 - Same as in Tract Orientation Mapping
 - Saves memory and enables wider generalization compared to using raw data
- **Output:** Replaces the Softmax layer with independent per-tract sigmoid outputs to account for crossing bundles

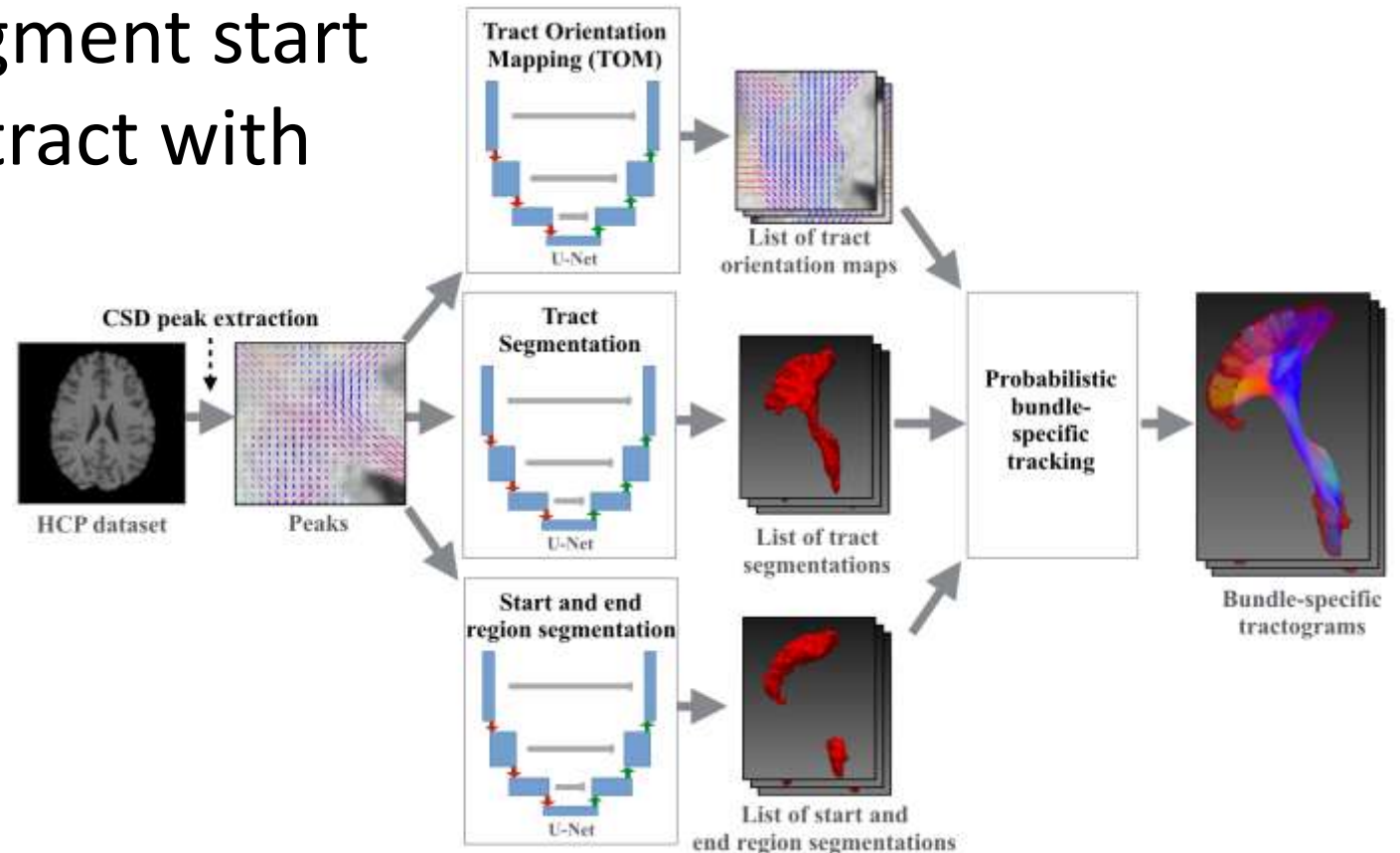
Handling 3D

- Like FastSurfer, TractSeg prefers 2D over 3D CNNs. However,
 - it trains *a single* 2D network for all slicing directions which is run three times on the full volume to achieve a segmentation
 - for segmentation (not for TOM), a *second-level network* is proposed for fusing the 2D segmentations



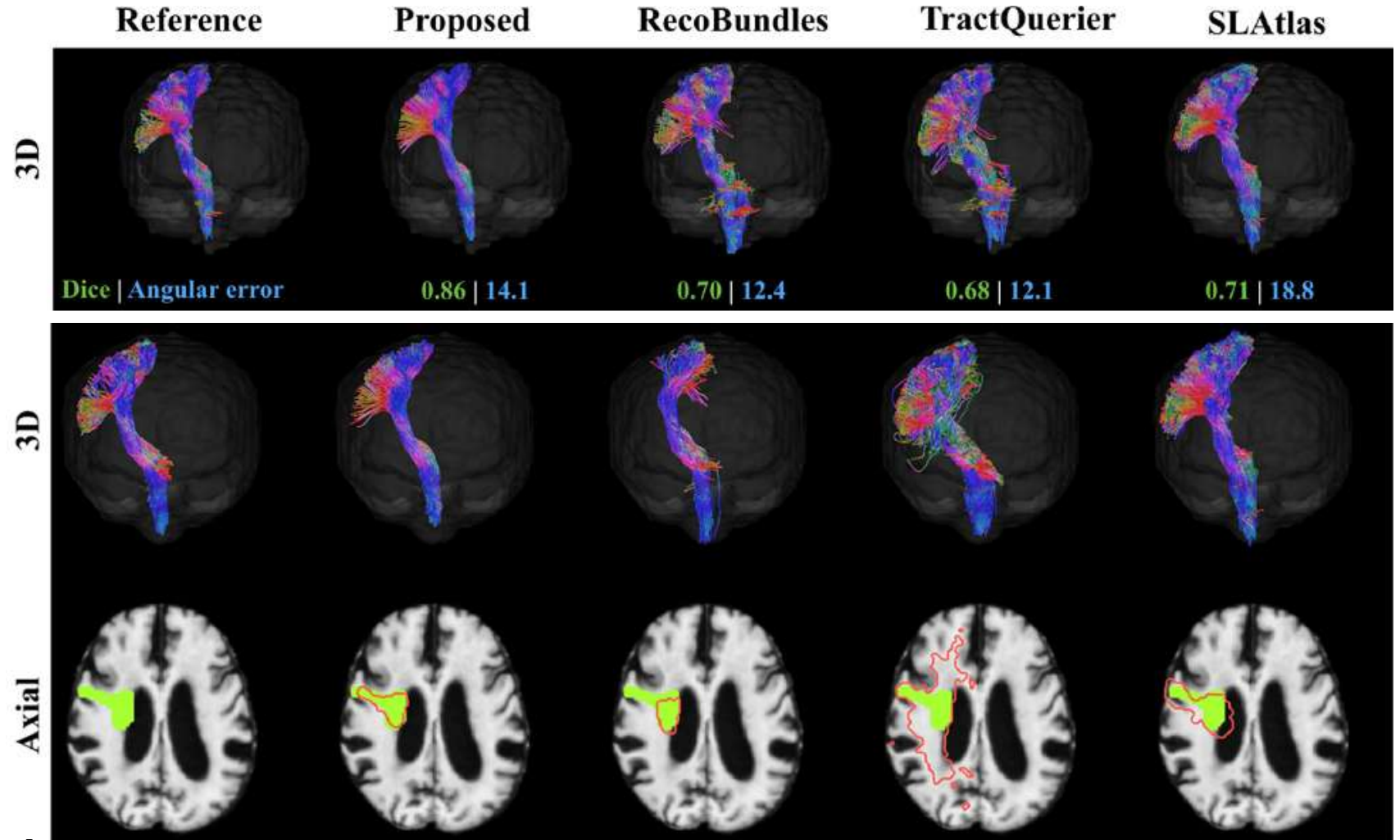
TOM-based Tractography: Full Pipeline

- **Problem:** Despite the segmentation mask, streamlines that end prematurely remain in the final result
- **Remedy:** Additionally segment start and end regions of each tract with yet another CNN
 - Discard streamlines that do not start / end in the desired regions



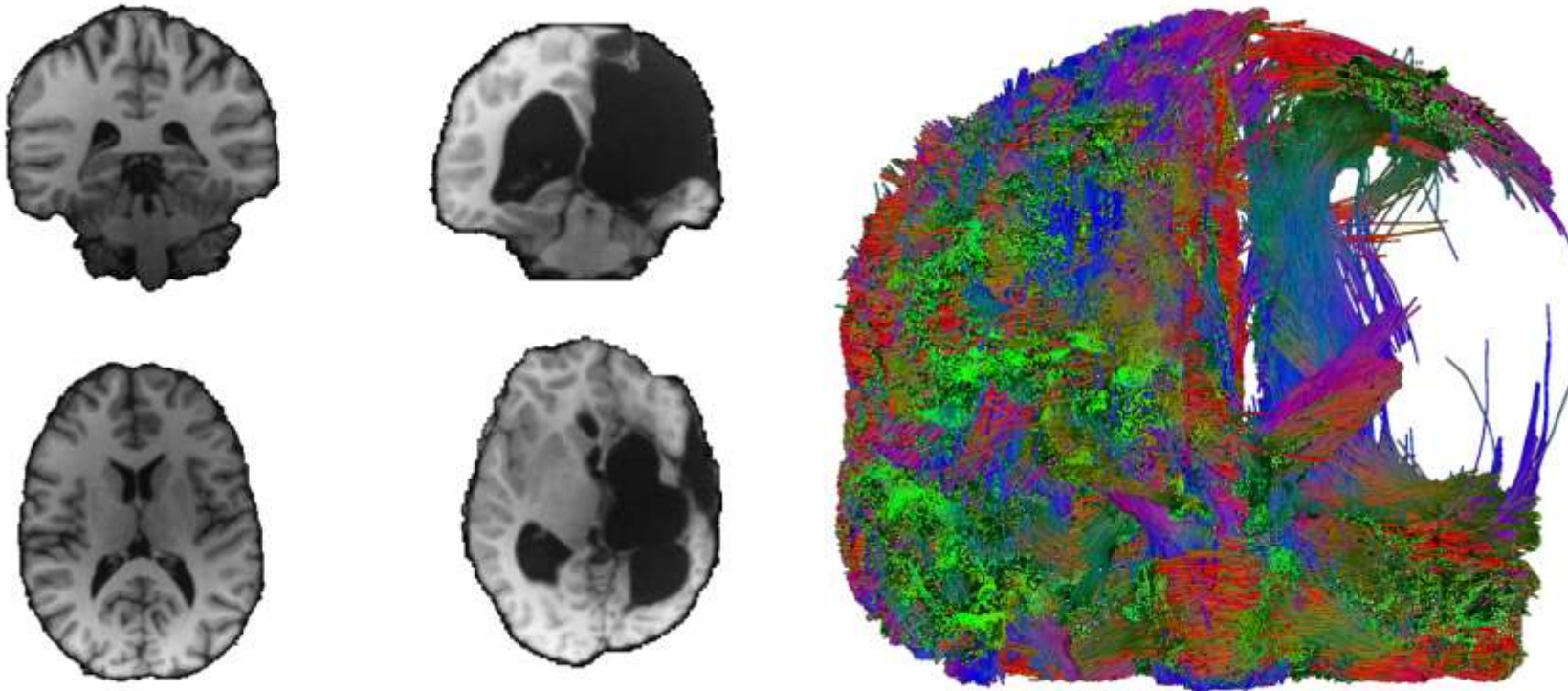
Results of TOM-based Tractography

- TOM-based tractography has produced useful results not just on healthy controls (top), but also for certain pathological cases (e.g., Alzheimer's, bottom)



Background: Hemispherotomy

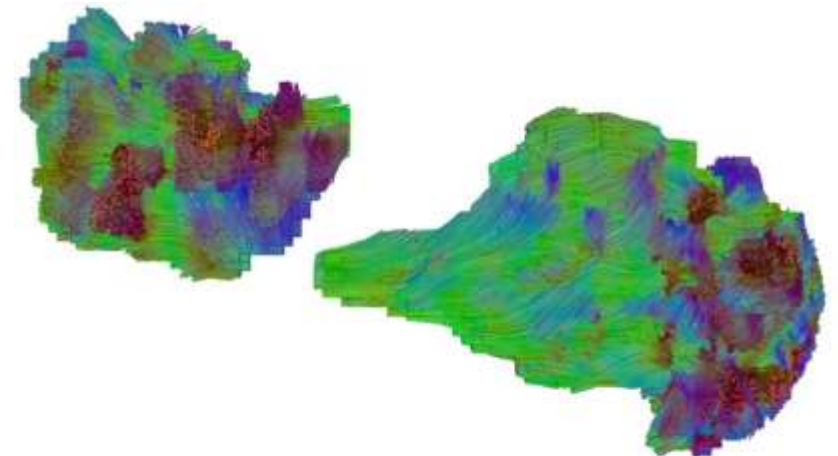
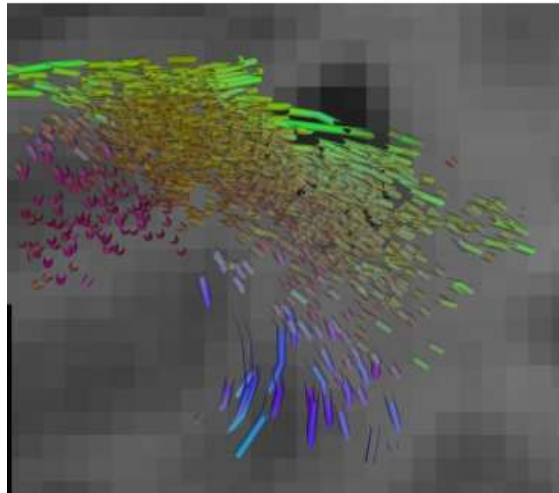
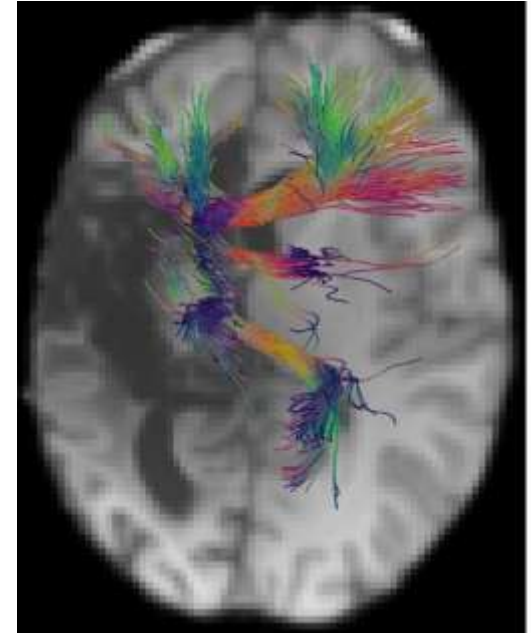
- Hemispherotomy is a last resort surgical intervention in certain types of epilepsy
 - Entirely disconnects one hemisphere from the other and the body



Healthy controls Hemispherotomy Whole brain post-hemispherotomy

Failure Modes of TOM in Pathological Data

- Applied TOM in 25 patients after hemispherotomy
- In several cases, TOM
 - „hallucinated“ parts of the corpus callosum, which are known to have been disconnected
 - continued tracts through obvious lesions
 - failed to reconstruct tracts due to deformations



Fiber Estimation as Low-Rank Approximation

- Traditional CSD uses fODF peaks as tracking directions
- Low-rank approximation framework [Schultz/Seidel 2008]
 - Optimizes a mixture of fibers that best explain the fODF
 - Expresses fODF as a higher-order tensor $\mathcal{T}$
 - Rank- r approximation of that tensor yields mixing weights $\lambda_i \in \mathbb{R}_+$ and directions $\mathbf{v}_i \in S^2$ of r fiber compartment:

$$\min_{\lambda_i, \mathbf{v}_i} \left\| \mathcal{T} - \sum_{i=1}^r \lambda_i \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \right\|^2$$

Anatomically Regularized Low-Rank Approximation

- Regularize low-rank approximation to include a fiber that is close to the TOM reference direction $\mathbf{v}_{TOM}$
 - Leave the corresponding volume fraction λ_i unconstrained

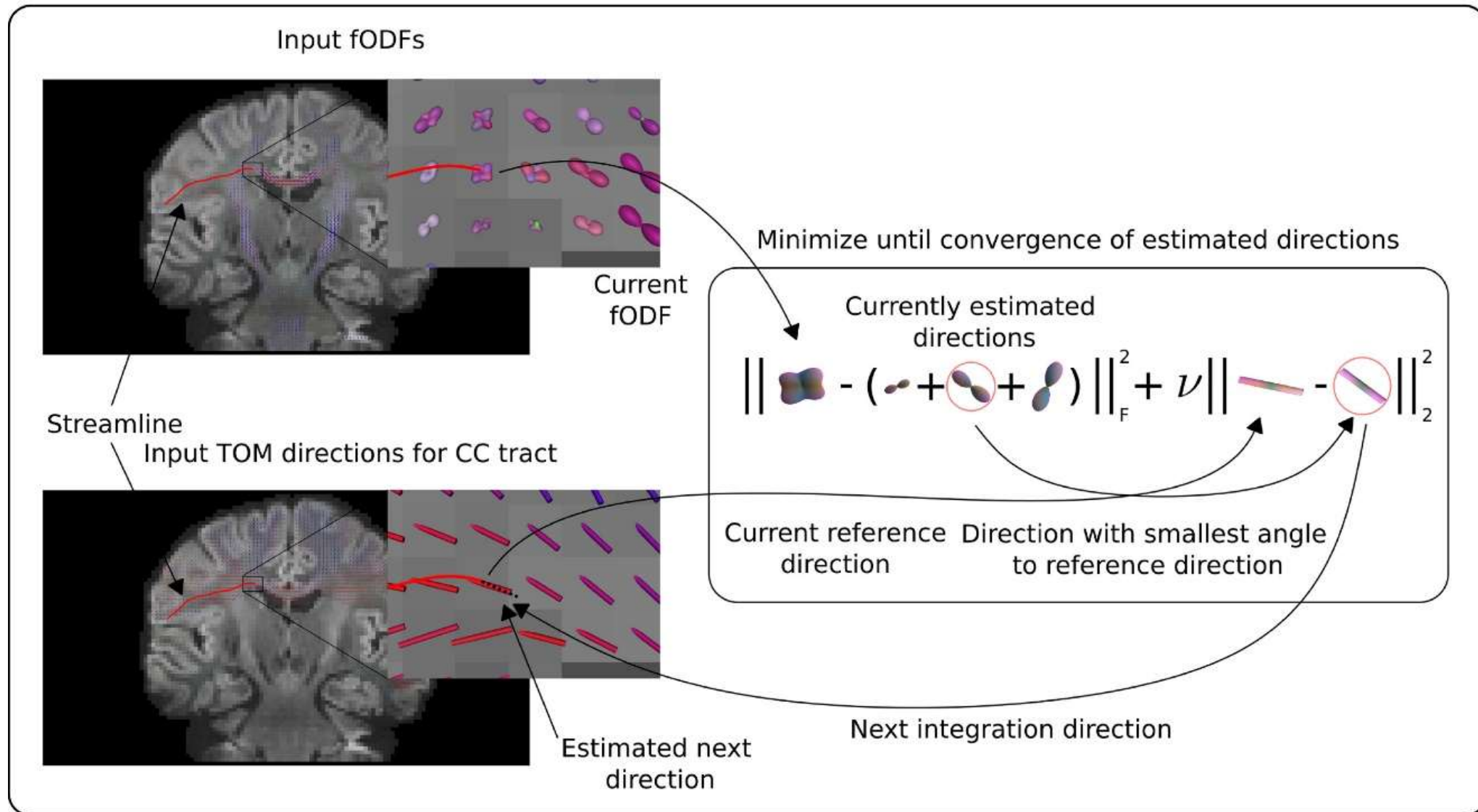
$$\left\| \sum_i \lambda_i \mathbf{v}_i^{\otimes 4} - \mathcal{T} \right\|_F^2 + \nu \left\| \mathbf{v}_{TOM} - \text{sgn}(\langle \mathbf{v}_{TOM}, \mathbf{v}_j \rangle) \mathbf{v}_j \right\|^2$$

with $j = \text{argmax}_i |\langle \mathbf{v}_{TOM}, \mathbf{v}_i \rangle|$

- Maintains **data fidelity**:
 1. Stop tracking when data does not support TOM direction
 2. Continue tracking outside TOM mask (without regularization) while the data supports it

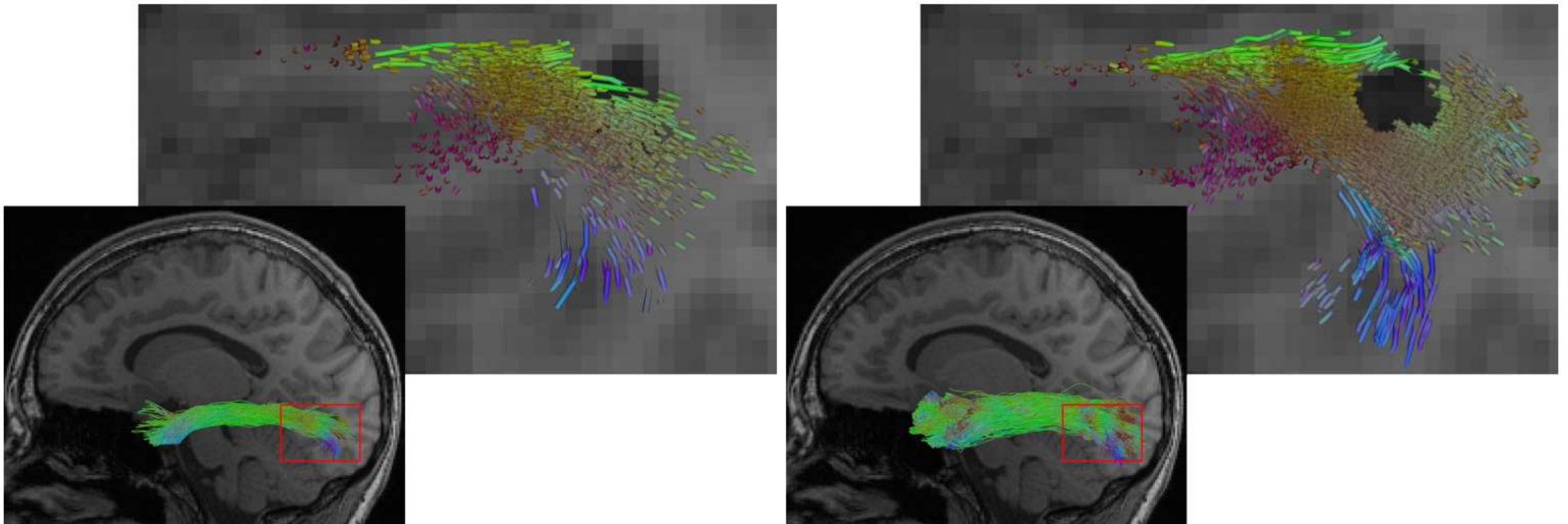
Illustration: TOM as Anatomical Regularizer

Tracking loop: Repeat until a stopping criterion is reached



Qualitative Results

- We obtain more complete tracts while avoiding implausible reconstructions, e.g., within lesions
 - Quantitative results: [Grün et al., NeuroImage:Clinical 2025]



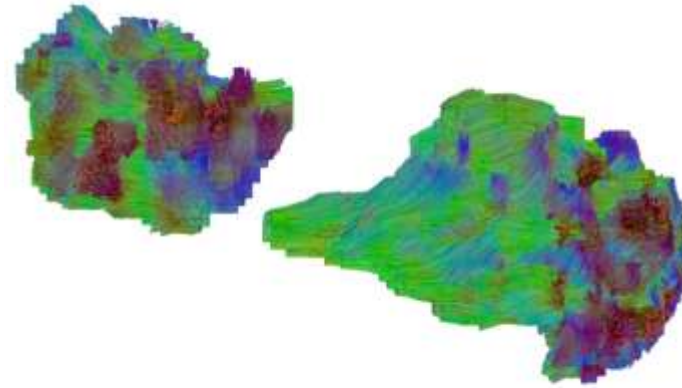
(a) TractSeg

(b) Regularized low-rank model

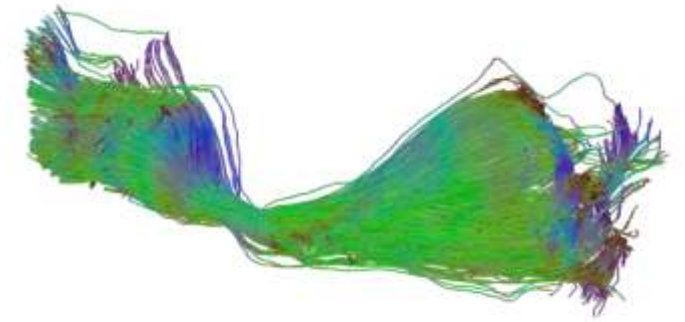
Further Qualitative Results

On patients and controls,

- we miss fewer tracts
- we obtain more complete reconstructions



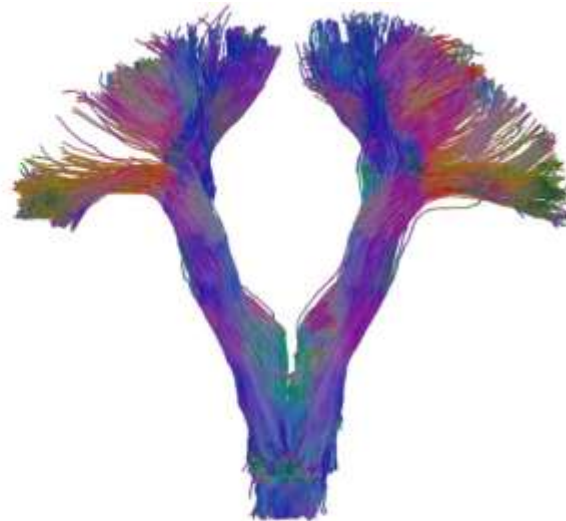
(a) TOM tractography (unfiltered)



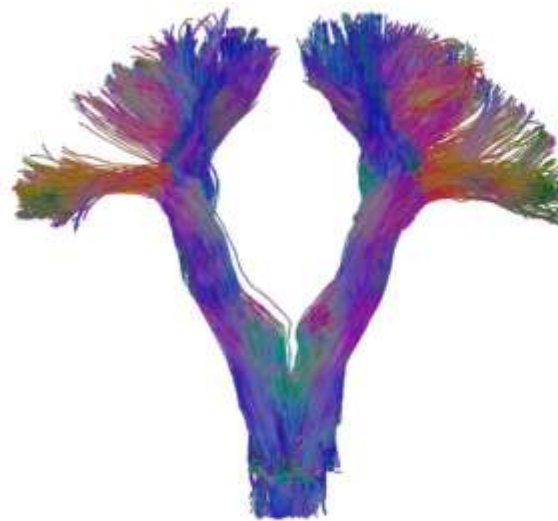
(b) Regularized low-rank model



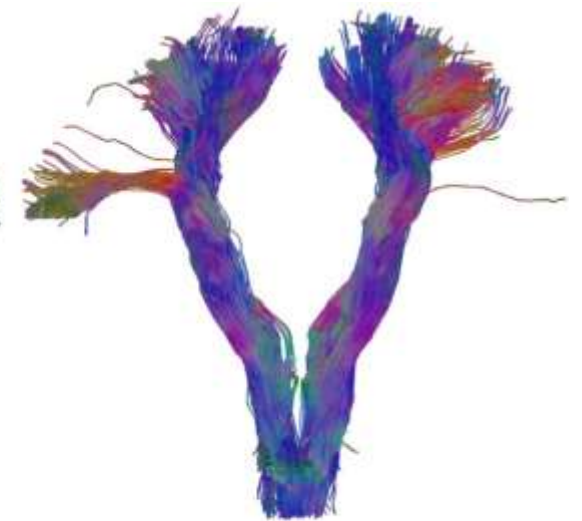
(a) TractSeg tractography



(b) Regularized with $\nu = 1.6$



(c) Regularized with $\nu = 0.8$



(d) Traditional low-rank

Summary: Learning-based Bundle-specific Tractography

- Mapping multi-vector fields from multi-fiber models to **tract orientation maps** resolves ambiguous correspondences
 - Requires anatomical knowledge that can be learned by a CNN
 - Only reconstructs specific tracts that were included in the training
- A state-of-the-art pipeline for **learning-based tractography** combines tract orientation mapping with
 - Explicit segmentation of the tracts' volumetric extents
 - Explicit segmentation of the tracts' start and end regions
- All building blocks use variants of the **U-Net** (Chapter 5.7)

Benefits and Limitations of Fiber Tracking

Benefits of Fiber Tracking:

- Successfully reconstructs many known bundles
 - Currently, only available method for in-vivo tract reconstruction
- Useful for neurosurgery and neuroscience
- Deterministic streamline-based methods are quite fast and simple

Limitations of Fiber Tracking:

- Streamlines **do not** have a direct anatomical counterpart
 - Individual axons are much smaller
 - Fiber bundles are not point-to-point connections
- Issues of validation and false positives

Summary: Diffusion MRI

- During its **spontaneous heat motion**, water interacts with tissue at a cellular level
- **Diffusion MRI** probes this restricted diffusion behavior to draw conclusions about structures that are far below image resolution
 - **Scalar indices** of tissue microstructure (e.g., Fractional Anisotropy)
 - Statistical analysis often involves skeletonization
 - **Tractography**: Reconstruction of white matter pathways
 - Multi-fiber models to track through crossings
 - Learning-based approaches like Tract Orientation Mapping reduce ambiguity by accounting for anatomical prior knowledge

Further Reading

- D.K. Jones (editor): *Diffusion MRI: Theory, Methods, and Applications*. Oxford University Press, 2011
- Tournier, Mori, Leemans: *Diffusion Tensor Imaging and Beyond*. *Magnetic Resonance in Medicine* 65(6):1532-1556, 2011
- Thomas Schultz: *Neuroimage Acquisition and Analysis*. In: *Neurotechnology: methods, advances and applications*, pp. 37-67, IET 2020
- J Grün, T Bauer, T Rüber, T Schultz: *Deep learning based tractography with TractSeg in patients with hemispherotomy: Evaluation and refinement* *NeuroImage: Clinical* 45:103738, 2025

Exam-Like Questions

1. What could a reduced value of Fractional Anisotropy in a patient cohort with respect to a group of healthy controls indicate?
2. Briefly explain a difference between Voxel Based Morphometry and the statistical analysis that is commonly done in Diffusion Tensor Imaging.
3. Name a mathematical model for estimating multiple fiber directions per voxel in diffusion MRI. Explain why such models are needed.